

***Actinomyces* and *Mycobacterium* spp.**

MTMI 203

Medical Bacteriology and Mycology I

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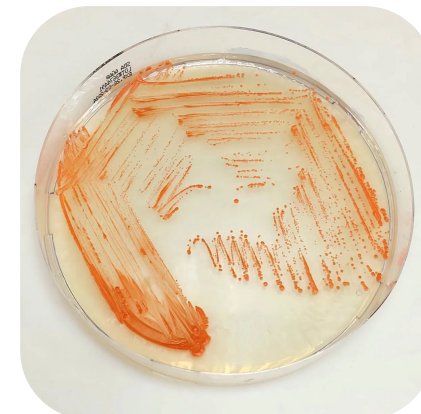
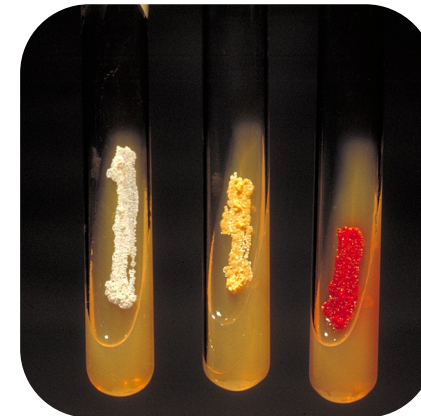
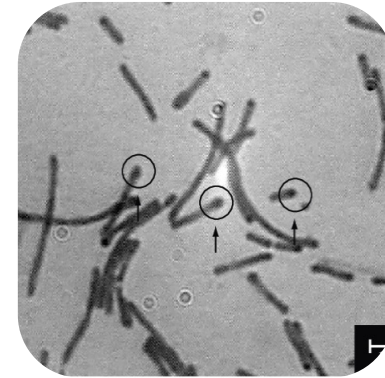
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Actinomyces – General Characteristics

- Belong to the phylum *Actinomycetota* (Actinobacteria)
- Gram-positive bacteria → *ท่อน้ำเงินยาว* rod coccoid form coccobacilli
- Filamentous growth : can fragment into rod or coccoid forms
- Colonies often exhibit diverse pigmentation
- High GC content
- Ability to produce a wide range of bioactive compounds
- Composed of several orders, such as Acidothermales, Actinomycetales, Frankiales, and **Mycobacteriales**.



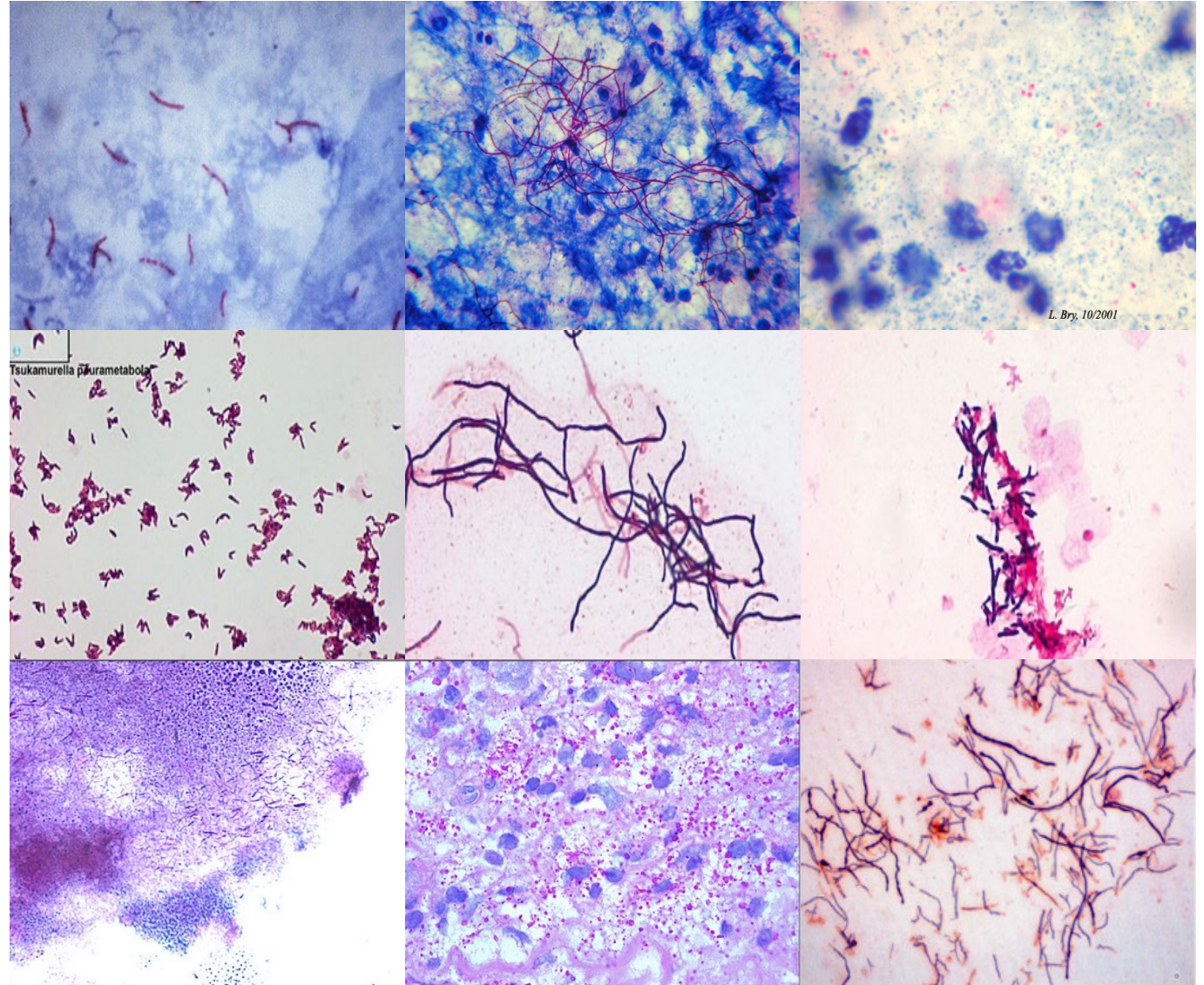


- Aerobic *Actinomycetes*

- *Mycobacterium* spp.
- *Nocardia* spp.
- *Rhodococcus* spp.
- *Gordonia* spp.
- *Tsukamurella* spp.
- *Dietzia* spp.
- *Streptomyces* spp.
- *Actinomadura* spp.
- *Tropheryma whipplei*

- Anaerobic *Actinomycetes*

- *Actinomyces* spp.

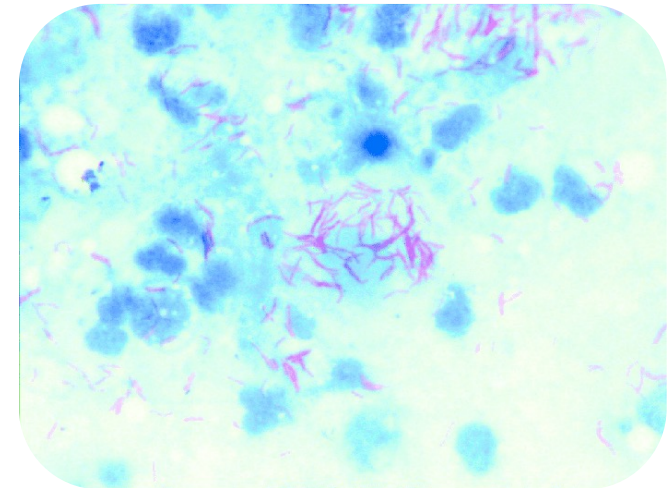
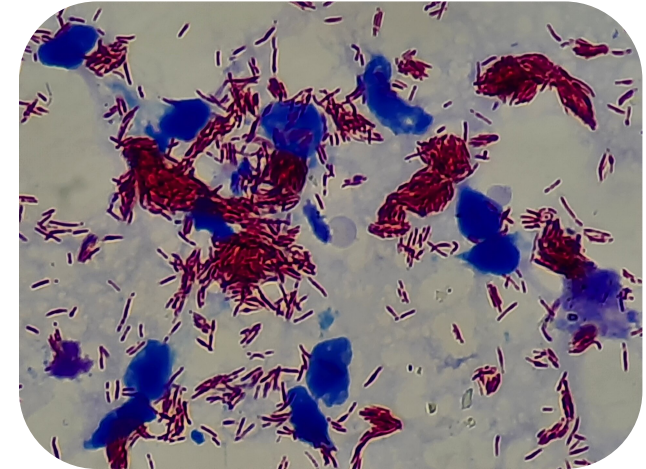




Mycobacterium spp.

- Belong to the family *Mycobacteriaceae*
- Intracellular and obligate aerobic bacteria
- Gram-positive, non-motile and non-spore forming bacteria
- Thin rod-shape bacilli (with size of $0.3 \times 3 \mu\text{m}$)
 - May present in pairs or as small clumps
 - May be present in slightly curved or straight rod
- Slow growing bacteria (~ 1 weeks – 8 weeks)
- Growth on artificial media
 - Except *M. leprae* and *M. lepromatosis*

→ คมไม่ชัด 94 จานเพาะ เลี้ยงเชื้อ





Mycobacterium spp.

- Lipid-rich cell wall (mycolic acid 70-90 atoms)

High lipid content (>60%)

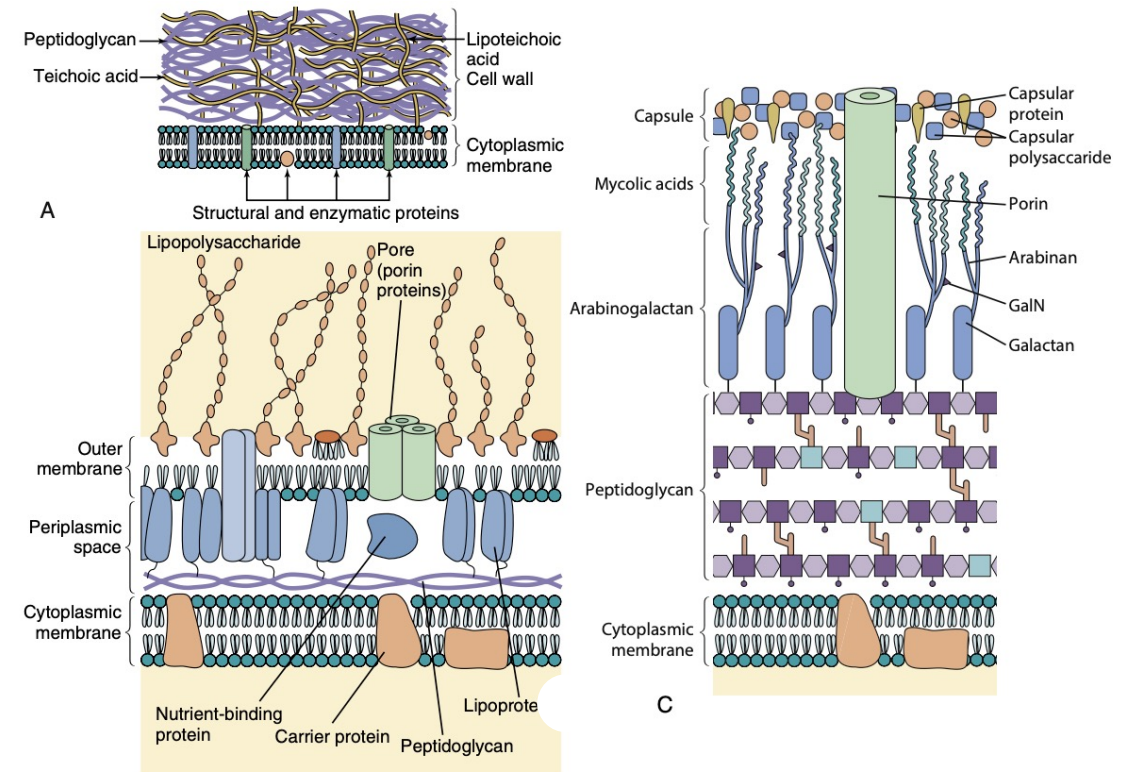
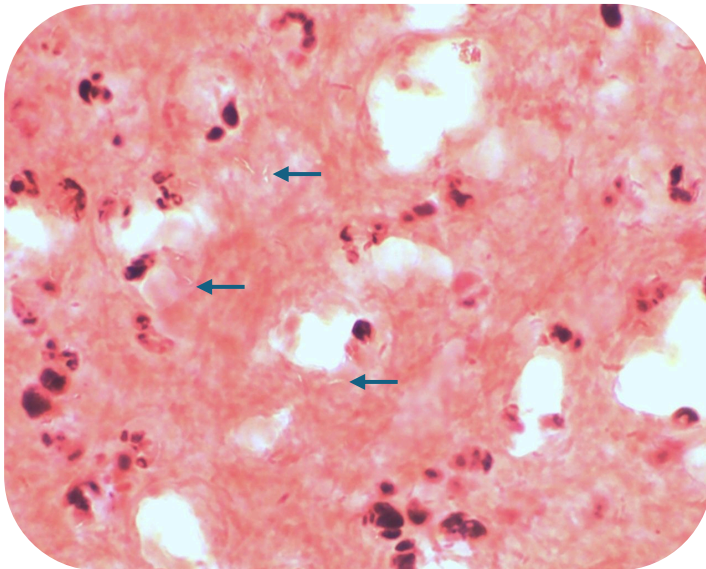
Slow growth

Resistant to detergents

Cannot stained by Gram's stain = “Ghost Cell”

Acid-fastness

* Resist to decolorization by acid alcohol
(3% HCl in 95% ethanol)





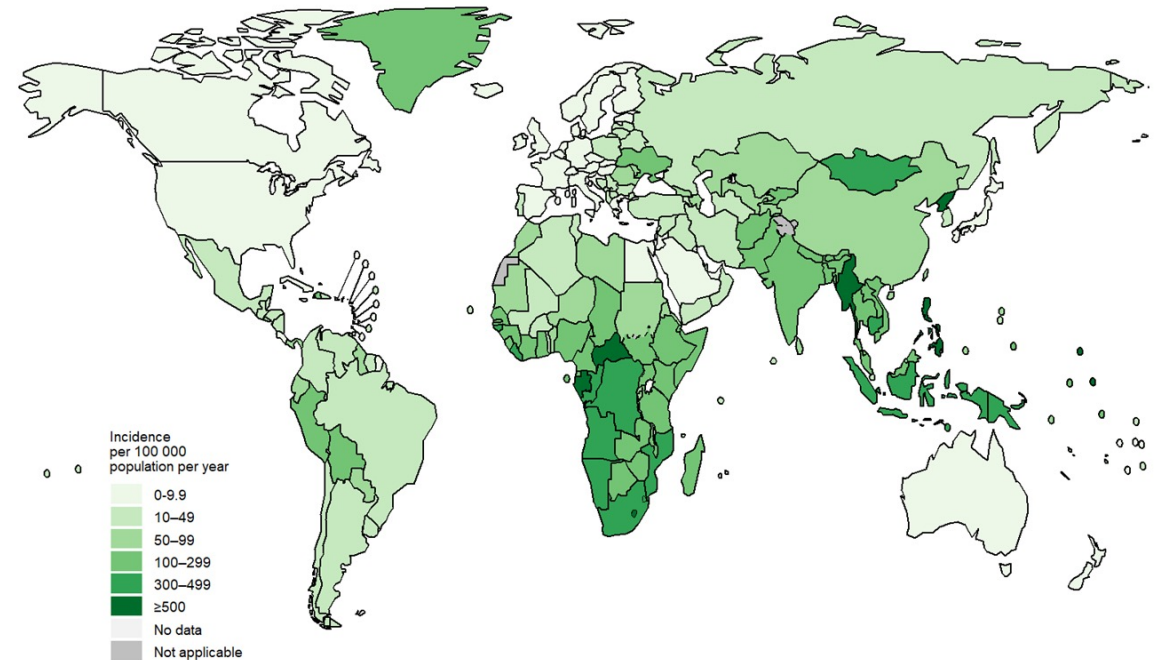
Classification

- Contain more than 190 species
- There are classified into 3 major groups based on molecular genetics
 - *Mycobacterium tuberculosis* complex (MTBC)
 - *Mycobacterium tuberculosis*
 - *Mycobacterium bovis*
 - *Mycobacterium africanum*
 - Nontuberculous Mycobacteria (NTM)
 - All other species not in MTBC and Lepromatous Mycobacterium
 - Lepromatous Mycobacteria
 - *Mycobacterium leprae* ①
 - lepromatosis ②



Mycobacterium tuberculosis complex

- Cause tuberculosis (TB) in both humans and animals
- 10.8 million new cases reported in 2023
- One third of the population is infected
- Most common in Southeast Asia, Africa and Eastern Europe
- **Strong** risk factor is associated with “HIV/AIDs patients” and “immunocompromised”





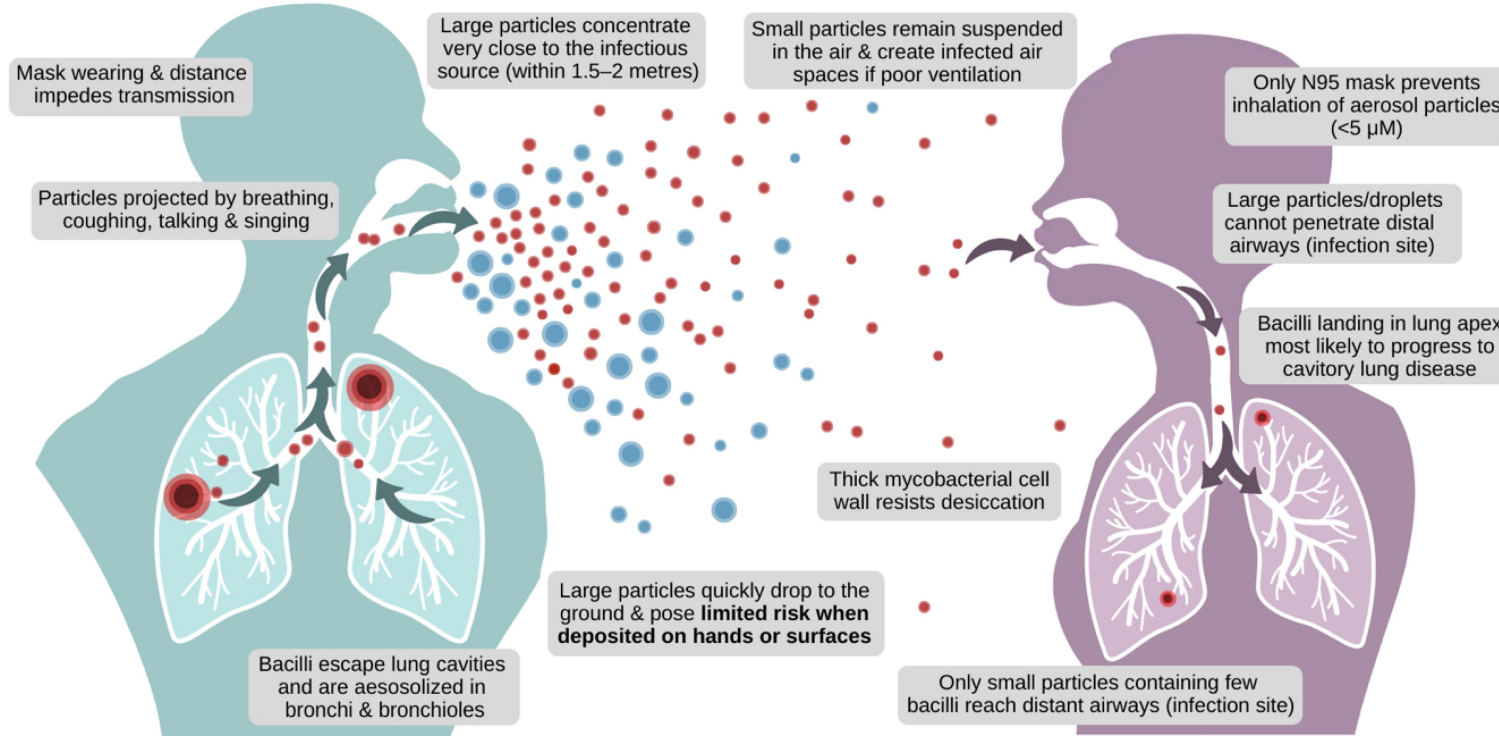
Important pathogenic species

- Human pathogens
 - *M. tuberculosis*: most common
 - *M. africanum*: restricted to West Africa
 - *M. canettii*: mainly in East Africa
- Zoonotic pathogens
 - *M. bovis*: cattle
 - *M. caprae*: goats, cattle
 - *M. microti*: rodents
 - *M. pinnipedii*: seals





Transmission



- Main route: Inhalation
- Infection site: small air sac (alveoli) in lung
- Infective dose: very low
(~ 10 acid-fast bacilli can cause TB infection)

Infectious Source

Ability to generate infectious aerosol
Bacillary load & tussive force
Potential asymptomatic transmission
(singing, talking, breathing)

Social

Number and duration of close contacts
Time spent in poorly ventilated spaces
Re-aerosolization after surface deposition (?)

Pathogen

Strain related variability; drug resistance
Viability/fitness/virulence of bacilli
Ability to withstand desiccation / UV light exposure

Environment

Ventilation – air exchange cycles/hour
Air pollution – increased airway inflammation
UV light and humidity – viability of infectious particles

Susceptible Host

Risk of infection

Proximity and duration of contact with
infectious source/ infected airspaces

Risk of Disease

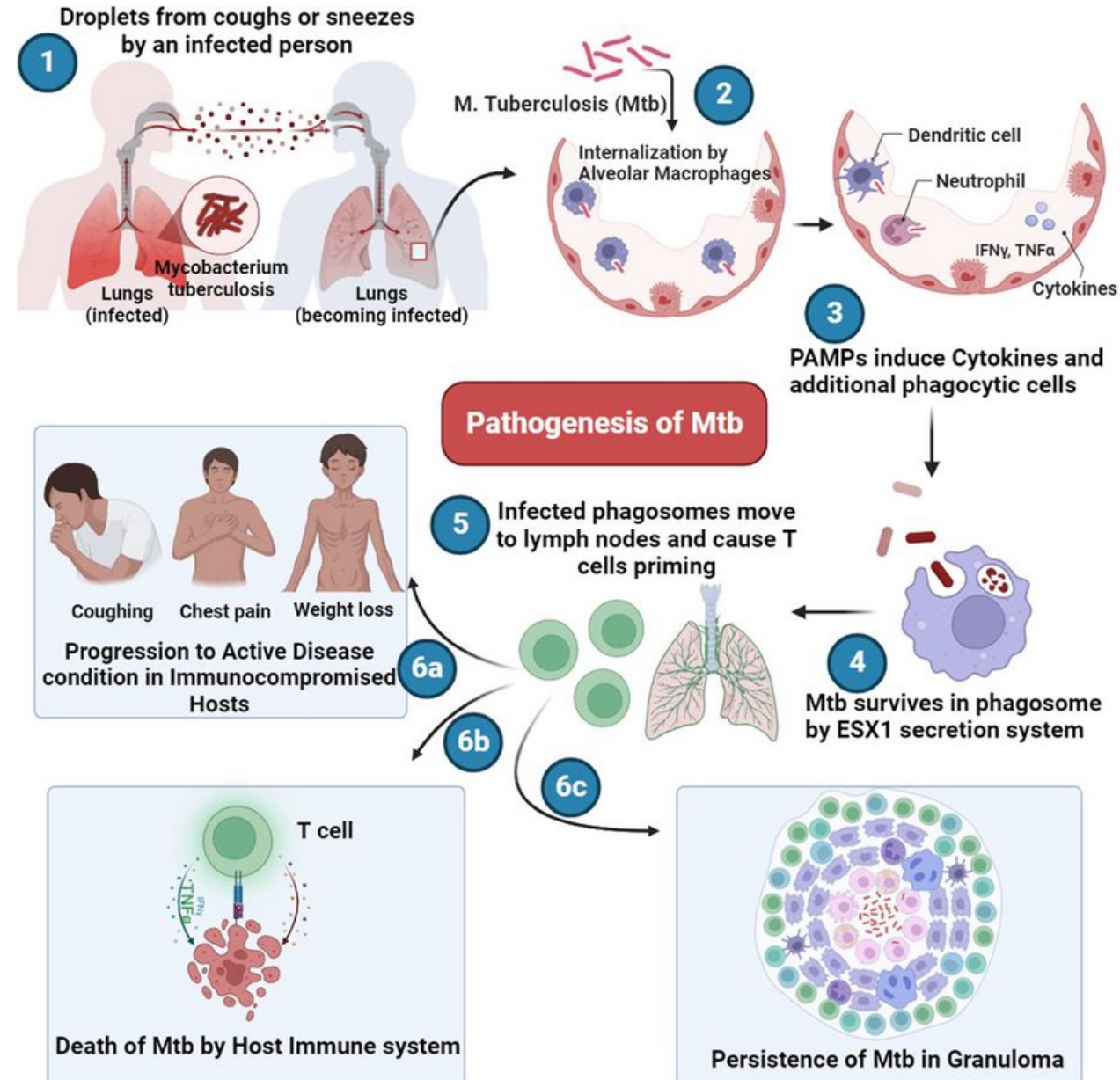
Systemic vulnerability – HIV/AIDS, young
age (immune immaturity), other T-cell
immune compromise
Lung vulnerability – structural lung damage
& airway inflammation



Mycobacterium tuberculosis complex

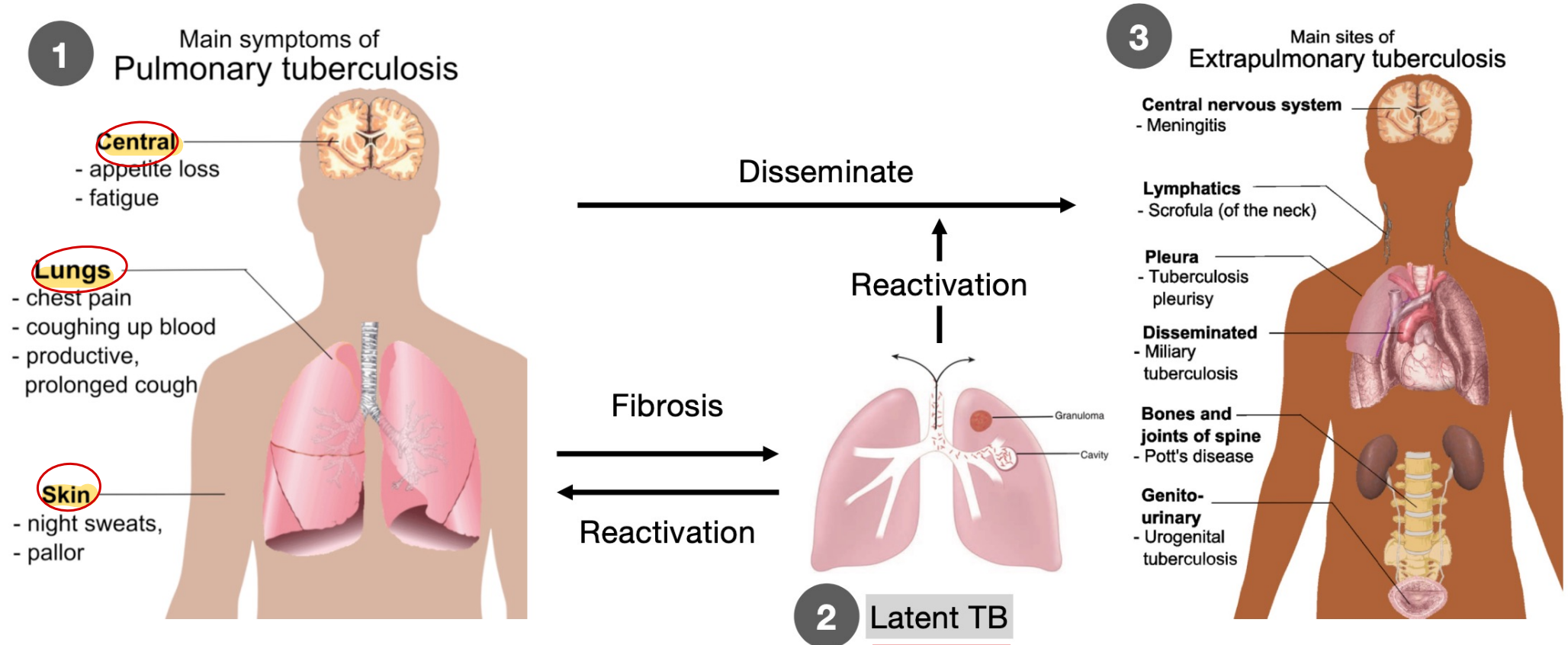
Pathogenesis

- Pulmonary TB





Pathogenesis





Virulence factor

- **Cord factor (Trehalose Dimycolate)**
 - A glycolipid on the cell surface that causes bacteria to grow in serpentine cord-like formations
 - **Inhibiting macrophage activation**
 - **Inducing granuloma formation** and caseation
 - **Triggering production of pro-inflammatory cytokines**
 - **Preventing phagocytosis**





Treatment

- The World Health Organization (WHO) recommends a 6-month regimen

Intensive Phase (First 2 Months)

- Drugs: Isoniazid (INH), Rifampin (RIF), Pyrazinamide (PZA), Ethambutol (EMB)
- Goal: Rapidly kill actively replicating bacteria and reduce bacterial load

Continuation Phase (Next 4 Months)

- Drugs: Isoniazid (INH) and Rifampin (RIF)
- Goal: Eliminate residual bacteria to prevent relapse



Mycobacterium tuberculosis complex

Treatment

- **First-line drug**

- Isoniazid (INH)
- Rifampin (RIF)
- Pyrazinamide (PZA)
- Ethambutol (EMB)



Critical group

→ **Multidrug-resistant TB (MDR-TB)**

- **Second-line drug**

- Fluoroquinolones (e.g., levofloxacin, moxifloxacin)
- Aminoglycosides (e.g., amikacin, kanamycin)
- Oxazolidinones (e.g., linezolid)
- Polypeptides (e.g. capreomycin)
- Cycloserine
- Ethionamide



Nontuberculous Mycobacteria (NTM)

- Diverse group of mycobacterial species (> 170 species)
- Widely distributed in the environment, including water, soil, and animals
- Some species were colonized on the skin, respiratory and gastrointestinal tract of healthy individuals
- Opportunistic pathogens
 - Spectrum of infection ranges from localized lesions to disseminated disease
- Route of infection: trauma, inhalation of infectious aerosols and ingestion
 - Not usually transmitted from person to person



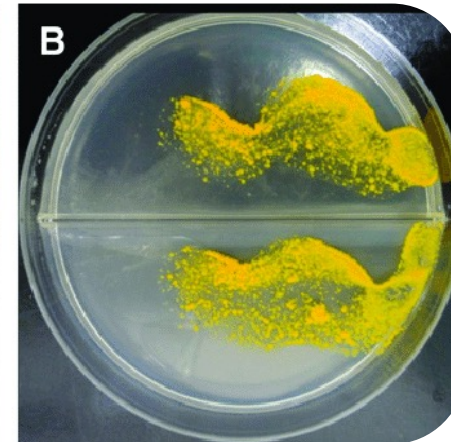
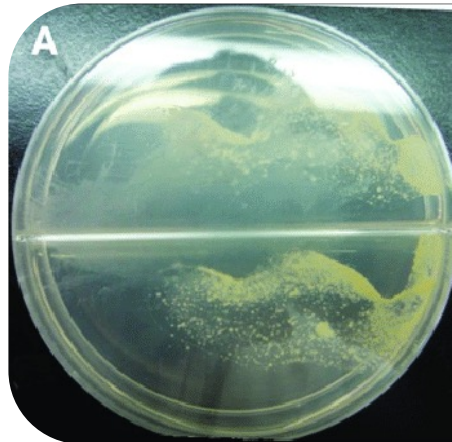
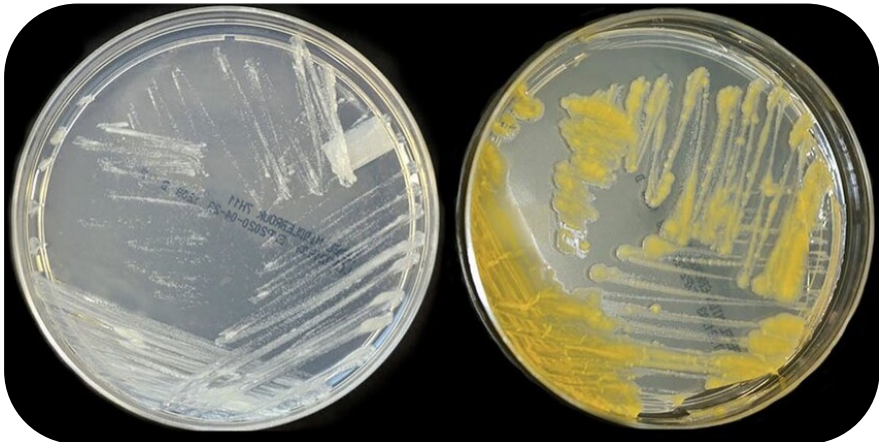
Nontuberculous Mycobacteria (NTM)

Classification based on pigment production (Runyon Classification)

- **Group I Photochromogens:**

Produce yellow-orange pigment only when exposed to light

- *M. kansasii*
- *M. marinum*
- *M. simae*





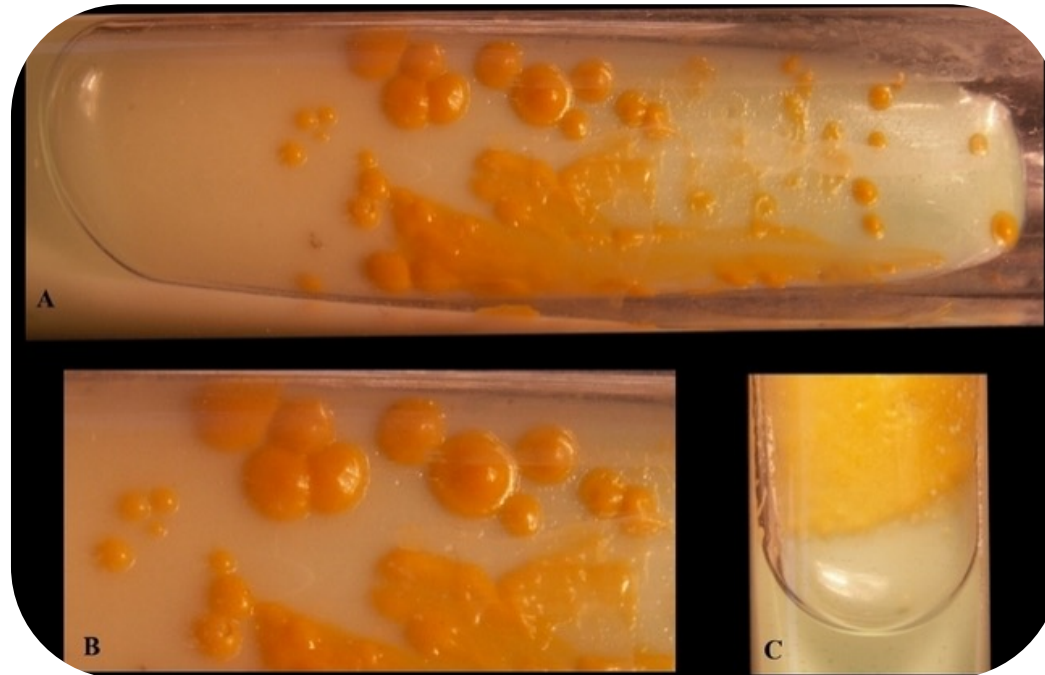
Nontuberculous Mycobacteria (NTM)

Classification based on pigment production (Runyon Classification)

- **Group II Scotochromogens:**

- Produce pigment in both light and dark conditions

- *M. scrofulaceum*
- *M. goodii*





Nontuberculous Mycobacteria (NTM)

Classification based on pigment production (Runyon Classification)

- **Group III Nonchromogens:**

No pigment production; colonies are white or pale

→ ขาว ใส

- *M. avium-intracellulare complex* (MAC)
- *M. ulcerans*

- **Group IV Rapid Growers:**

Not specifically classified by pigment production but form colonies quickly

Colony formation within 7 days

- *M. chelonae*
- *M. abscessus*
- *M. fortuitum*



Nontuberculous Mycobacteria (NTM)

Clinical manifestation

- Wide spectrum of clinical manifestations depending on the ^①species, ^②host immune status, and ^③site of infection

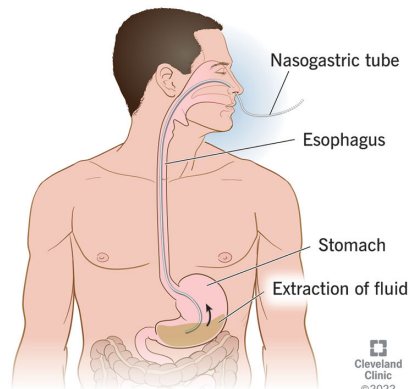
Disease ^③	Organism
Pulmonary infection	MAC, <i>M. xenopi</i> , <i>M. kansasii</i> , <i>M. abscessus</i>
Lymph node infection	MAC
Cutaneous infection	<i>M. chelonae</i> , <i>M. fortuitum</i> , <i>M. marinunnm</i> , <i>M. ulcerans</i>
Disseminated infection	MAC, <i>M. kansasii</i>
Catheter-related infection	Rapid Growers group



Clinical specimens

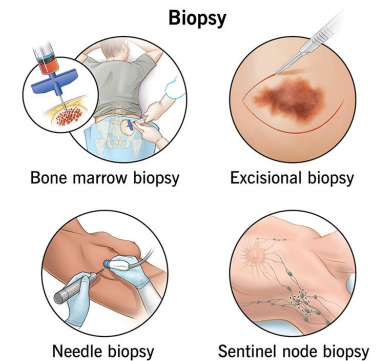
- Pulmonary tuberculosis

- Sputum
- Bronchoalveolar lavage (BAL)
- Gastric lavage fluids



- Extrapulmonary tuberculosis

- Blood
- CSF
- Body fluids
- Tissue biopsy

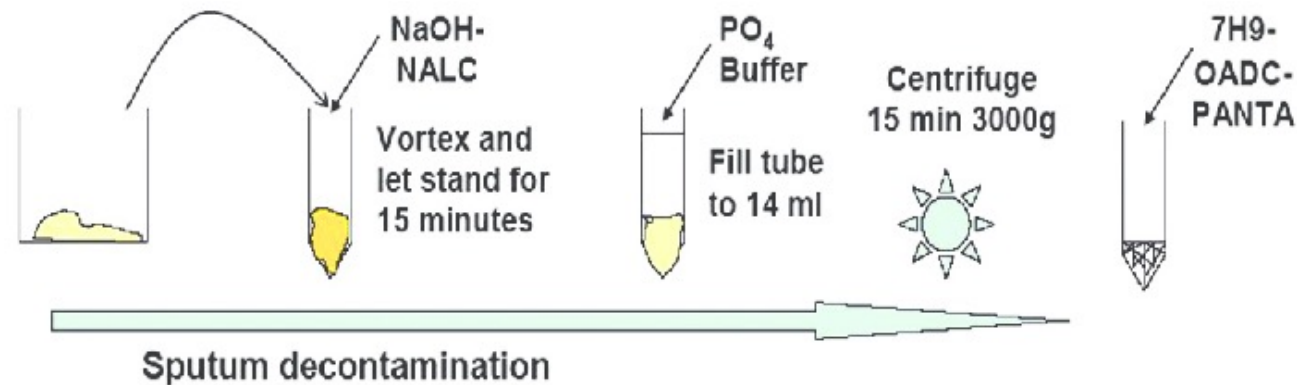




Specimen processing

- Decontamination + concentration process

- Eliminate contaminating flora, particularly from the upper respiratory tract
- Digesting organic debris and proteinaceous materials
- Most common method : NALC-NaOH
 - NALC (N-Acetyl-L-Cysteine) : mucolytic agent to liquefy the sputum
 - NaOH (Sodium hydroxide) : decontaminate the sample





Microscopic direct examination

- Acid-fast bacilli (AFB) stains
 - Ziehl-Neelsen staining -> Hot stain method
 - Kinyoun staining -> Cold stain method
- Fluorochrome stains
 - Auramine O



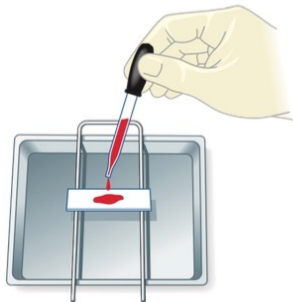
Microscopic direct examination

- Acid-fast bacilli stain



Ziehl-Neelsen Method

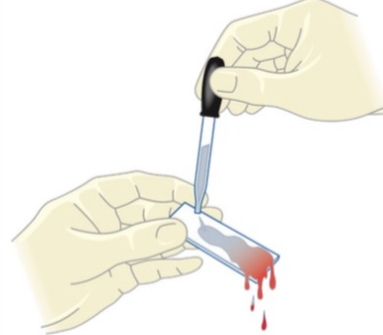
- 1a Heat method:** Apply carbol fuchsin and steam over a beaker of boiling water that is placed on a hot plate for 5 minutes. Do not allow the stain to evaporate.



Kinyoun Method

- 1b Heatless method:** Apply carbol fuchsin with Tergitol for 5 to 10 minutes.

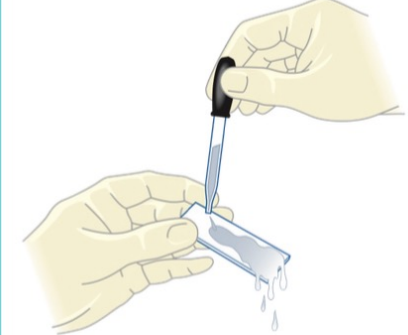
การหมักด้วยน้ำร้อน
เพื่อให้น้ำซึมเข้าเซลล์



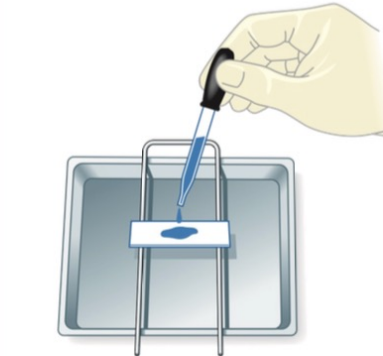
- 2** Cool and wash off stain with tap water.



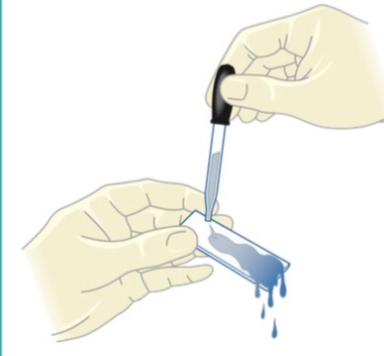
- 3** Add acid-alcohol drop by drop until the alcohol runs almost clear.



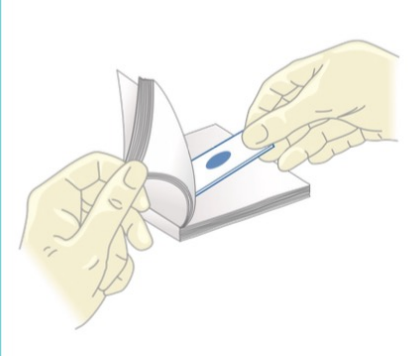
- 4** Wash off the acid-alcohol with tap water.



- 5** Counterstain with methylene blue for 2 minutes.



- 6** Wash off the methylene blue with tap water.

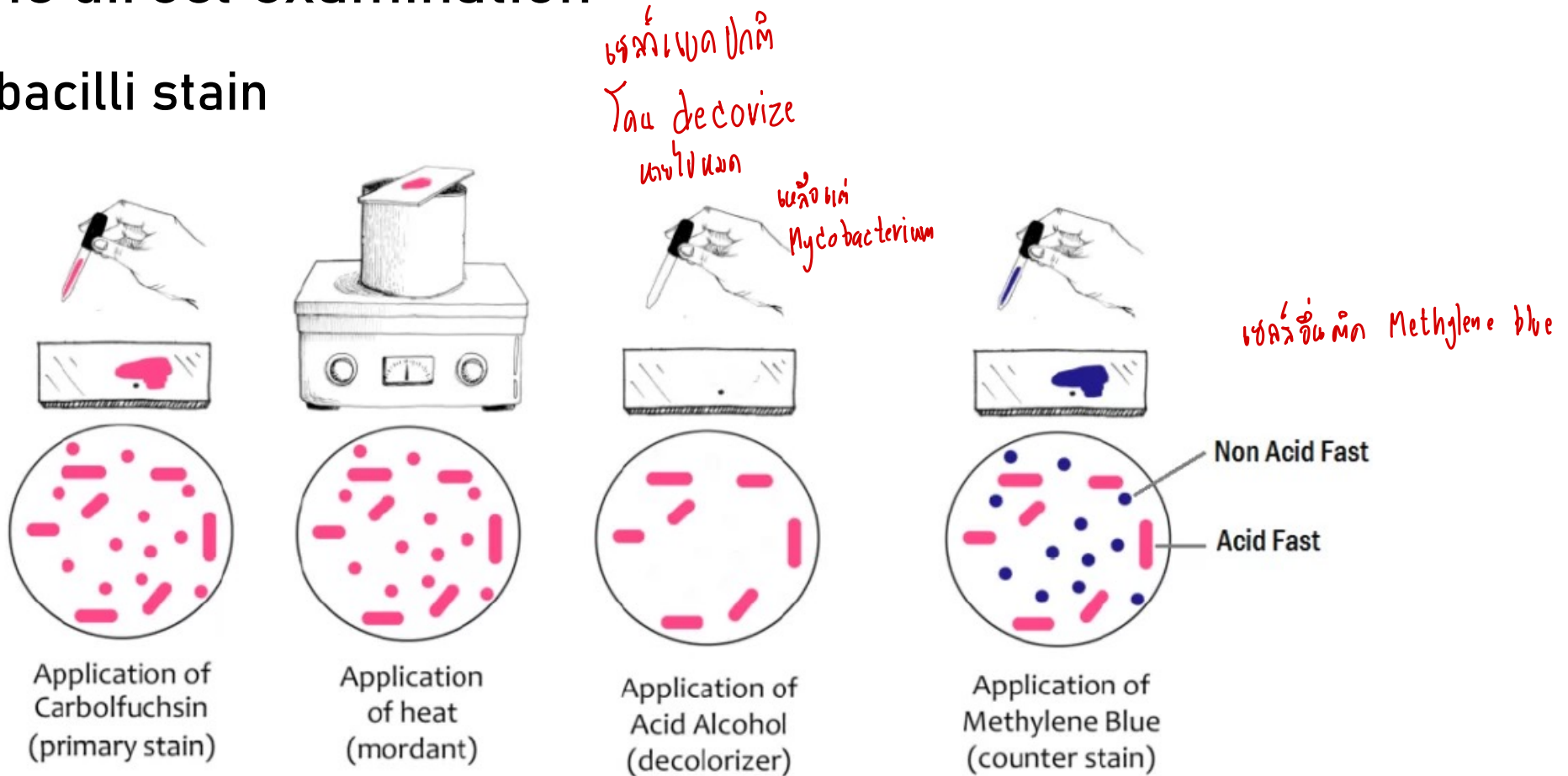


- 7** Blot the slide dry with bibulous paper.



Microscopic direct examination

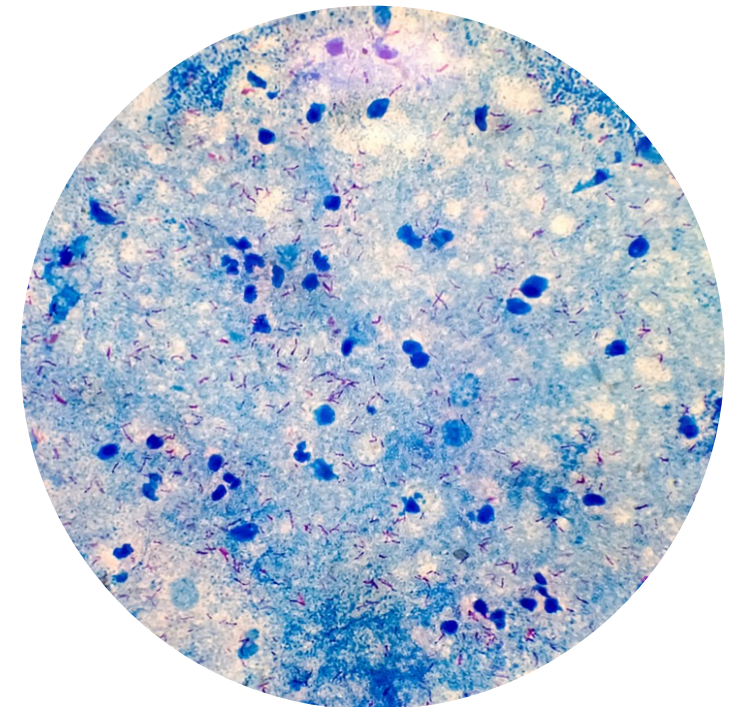
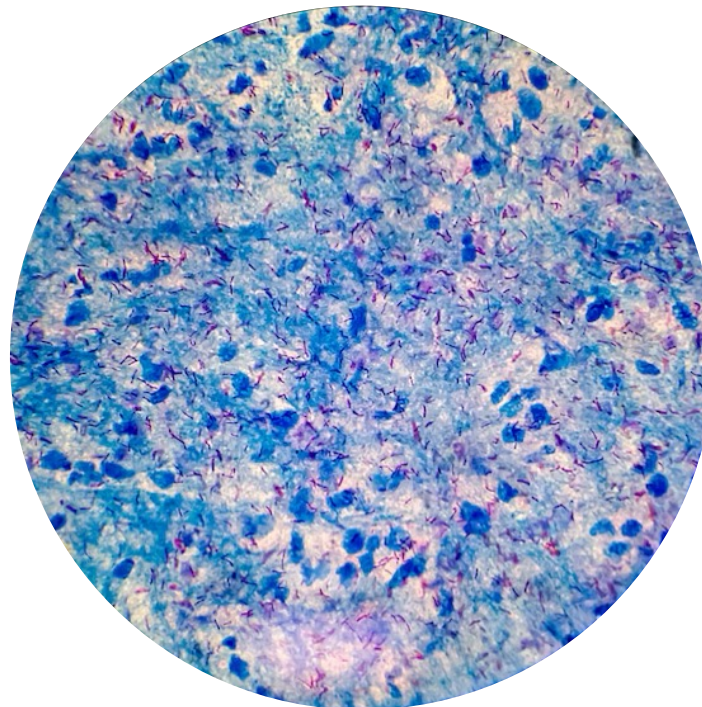
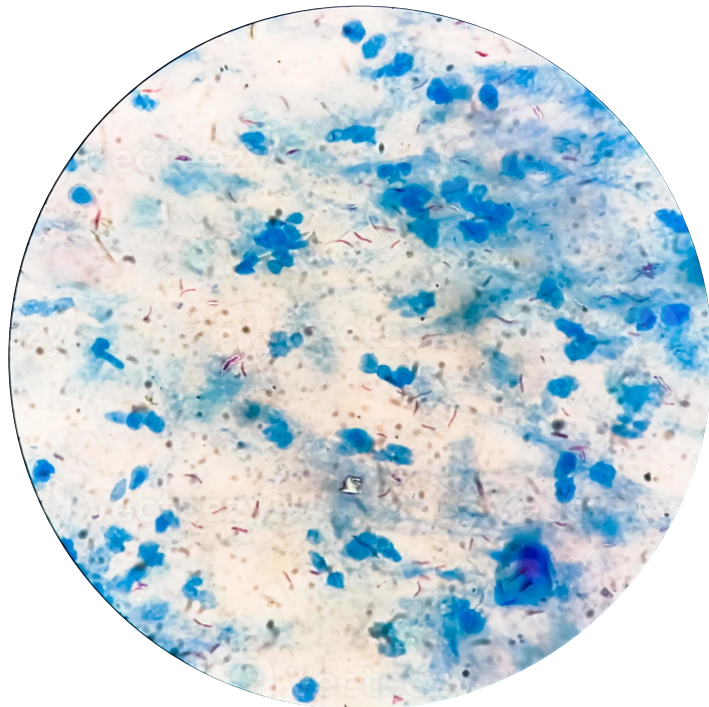
- Acid-fast bacilli stain





Microscopic direct examination

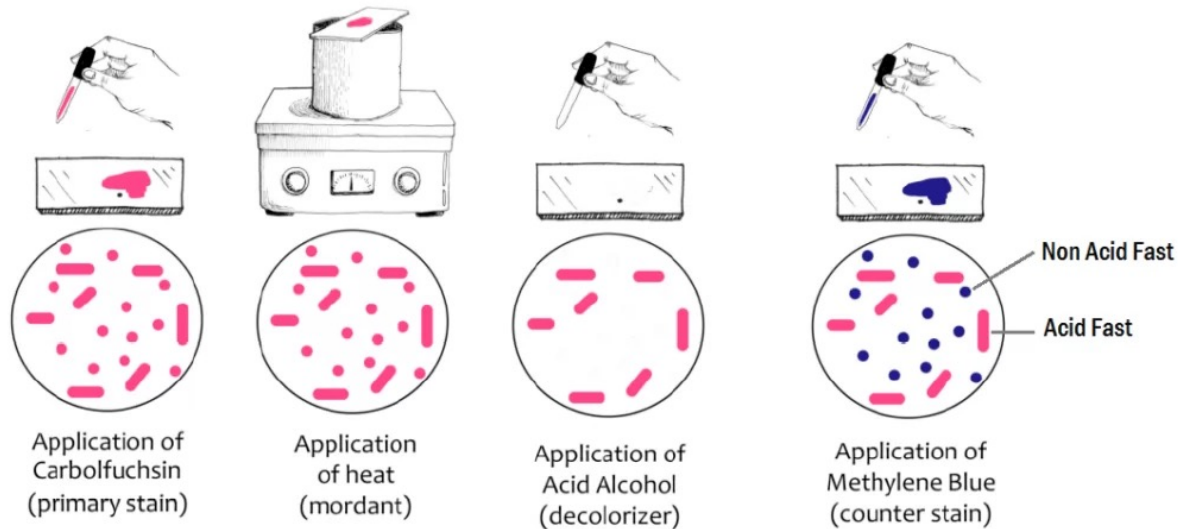
- Acid-fast bacilli stain
 - Limit of detection (LOD) = require at least 10,000 – 100,000 cells/mL in specimen



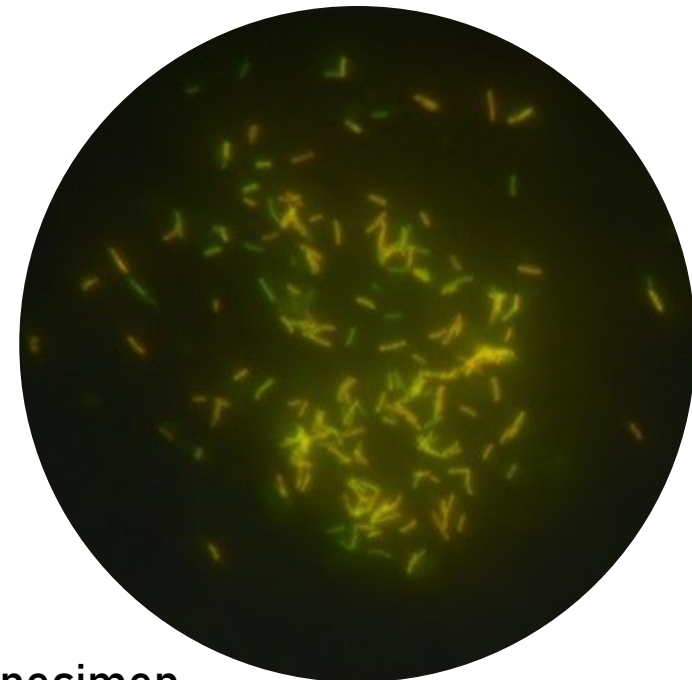


Microscopic direct examination

- Fluorochrome stain



- Primary stain : Auramine 0
- Decolorizer : Acid alcohol
- Counter stain : Potassium permanganate



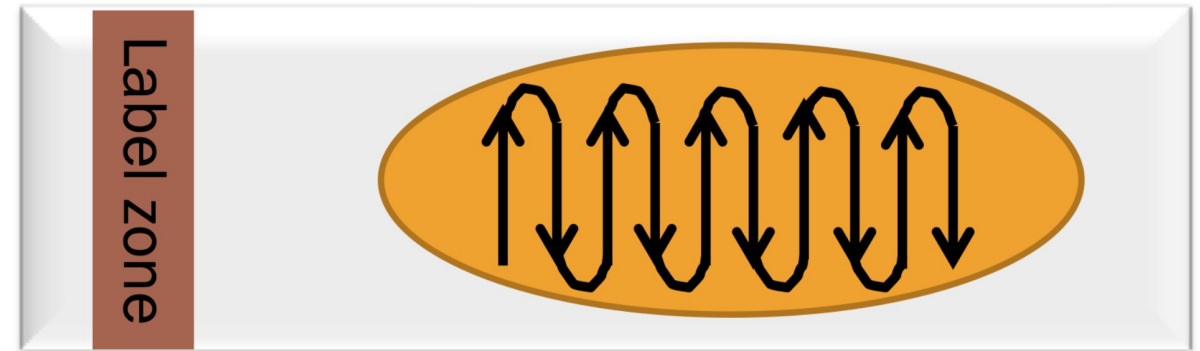
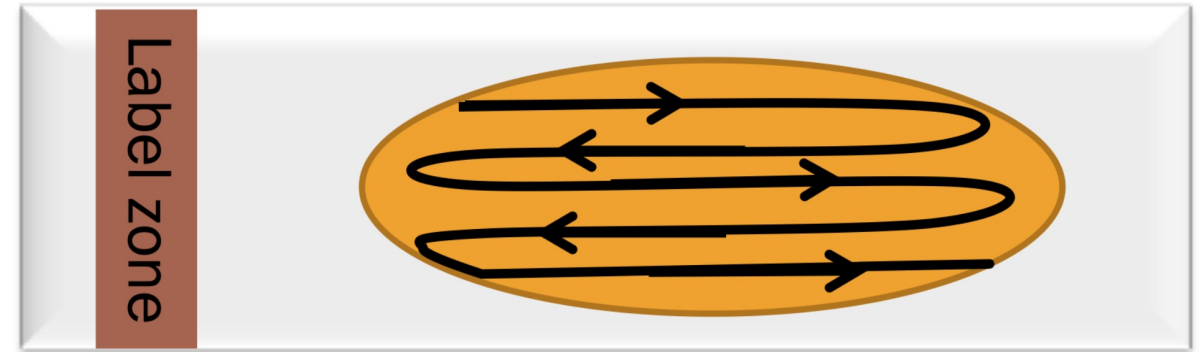
Limit of detection (LOD) = require at least 5,000 – 10,000 cells/mL in specimen



Microscopic direct examination

- Observation

- Zone of smear should be 1 x 2 inches
- Smear should not be too thick





Microscopic direct examination

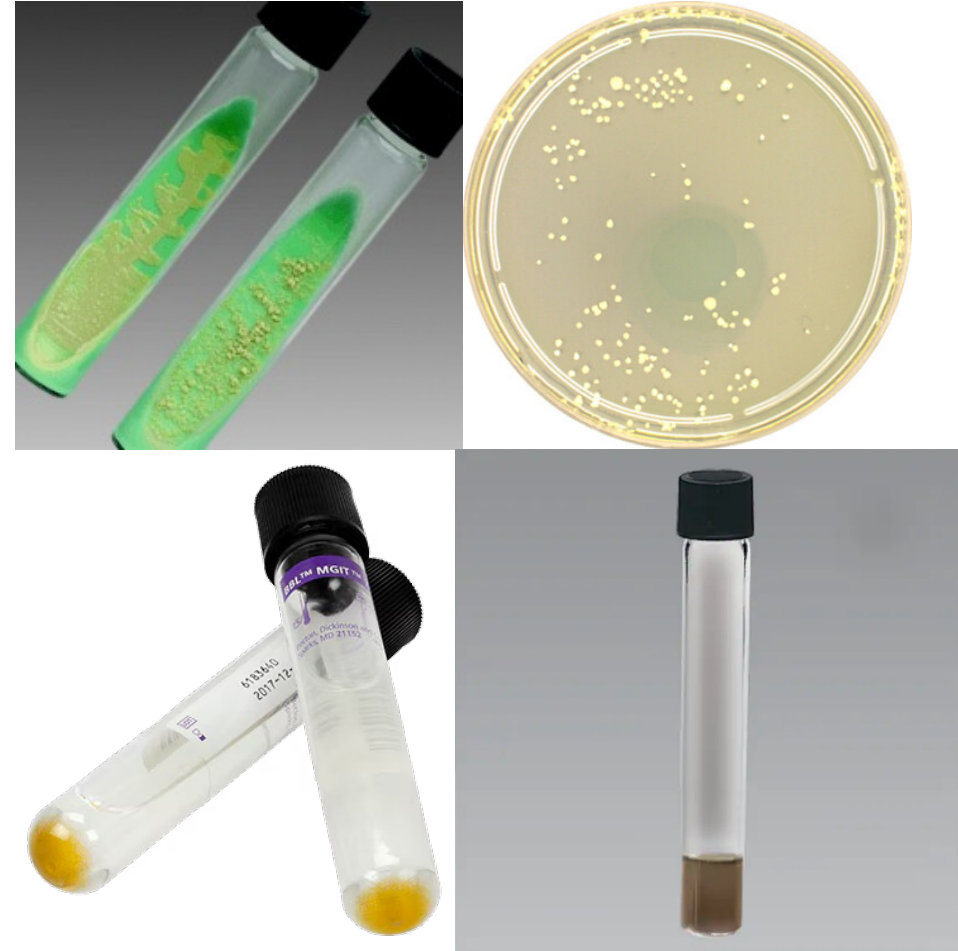
- Reporting system (WHO guideline)

Number of AFB found (1,000 X)	Results	Grading
0 cell / 100 oil fields (at least 100 fields)	Negative	No AFB seen
1-9 cells / 100 oil fields (at least 100 fields)	Scanty	Record exact number (e.g. AFB positive 3 cells)
10-99 cells / 100 oil fields (at least 100 fields)	Positive	AFB positive 1+
1-10 cells / 1 oil field (at least 50 fields)	Positive	AFB positive 2+
> 10 cells / 1 oil field (at least 20 fields)	Positive	AFB positive 3+



Culture

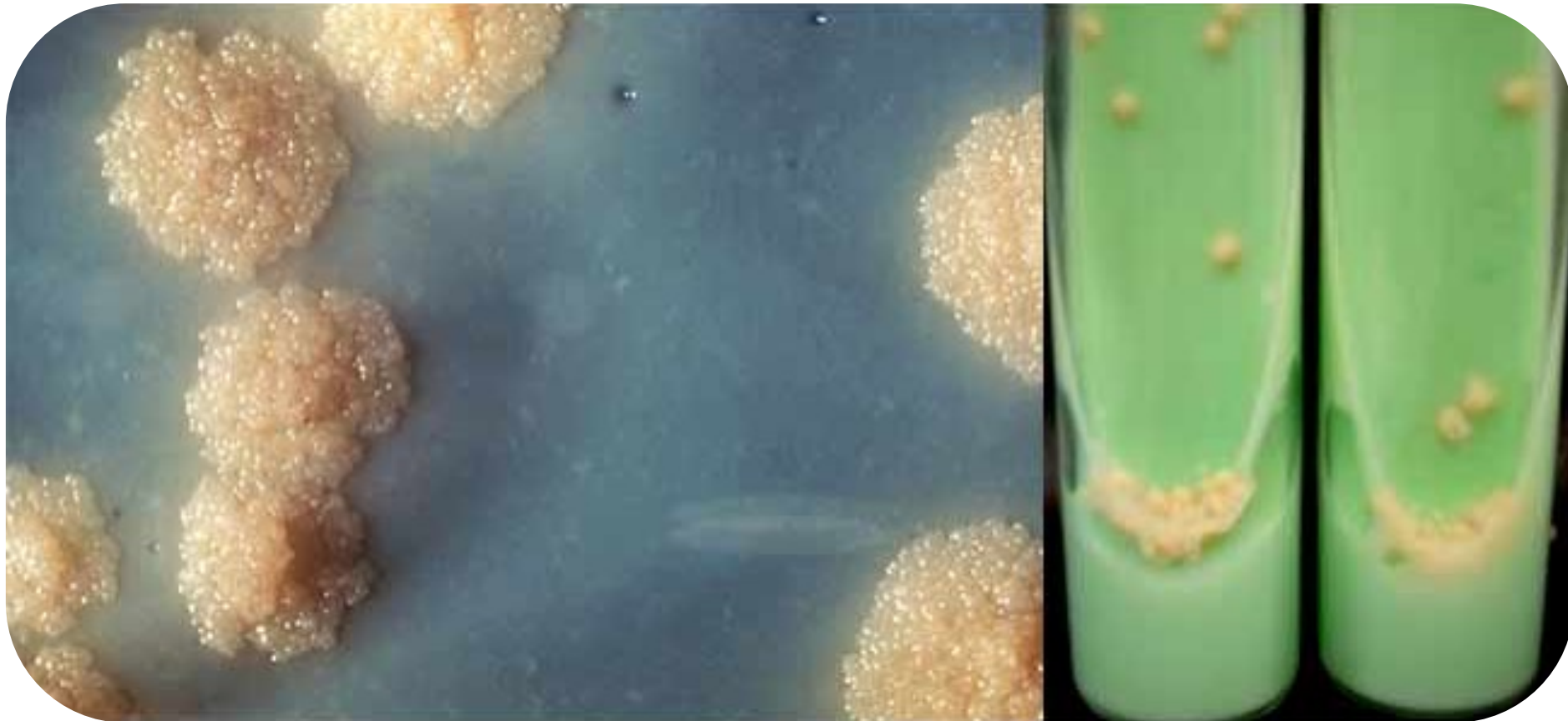
- Gold standard method
- Limit of detection = 10 – 100 cells
- 3 types of culture medium
 - Egg-based solid medium
 - Löwenstein-Jensen medium (LJ)
 - Ogawa medium
 - Agar-based solid medium
 - Middlebrook 7H10 and 7H11 medium
 - Agar-based liquid medium
 - Middlebrook 7H9
 - BACTEC 12B (MGIT)





Culture

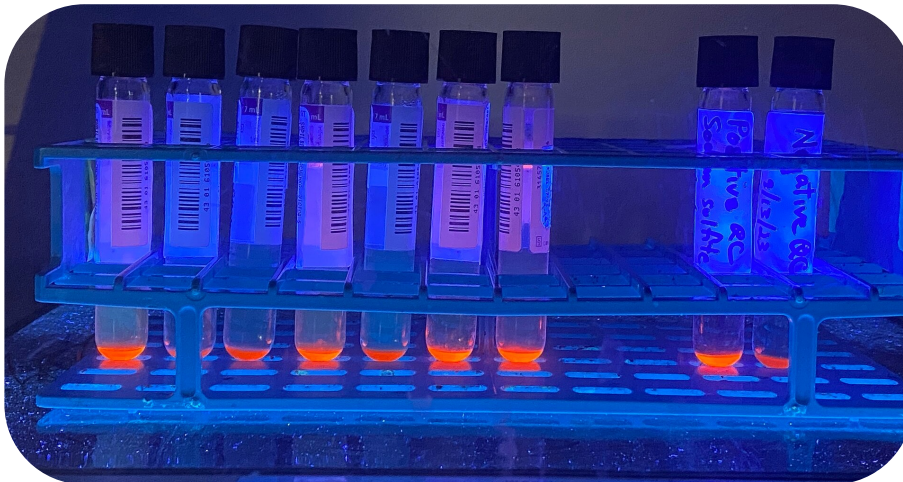
- Colony morphology of MTBC : Cauliflower like colony





Culture

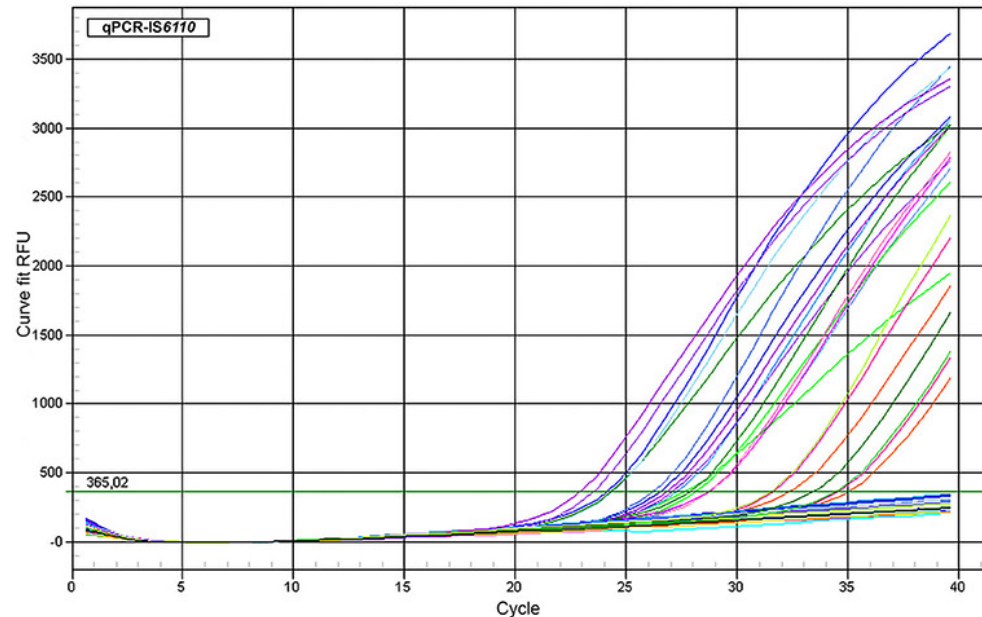
- Automated method
 - Mycobacteria Growth Indicator Tube (MGIT)
 - Modified Middlebrook 7H9 broth
 - Fluorescence-quenching-based oxygen sensor
 - More oxygen consumption increase fluorescence : indicate microbial growth





Molecular diagnostic method

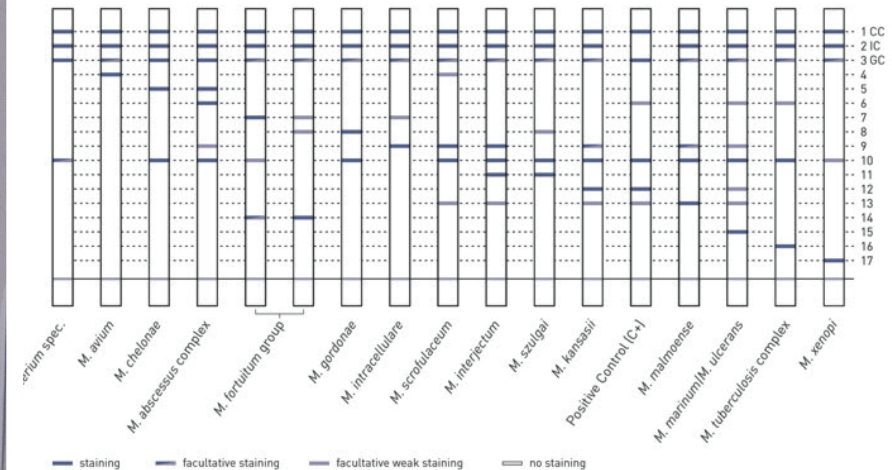
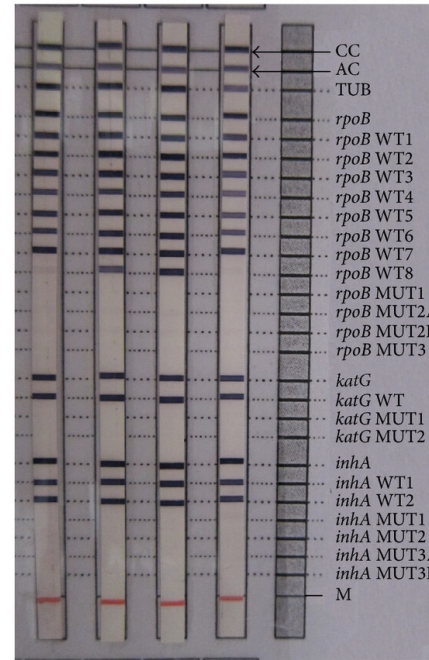
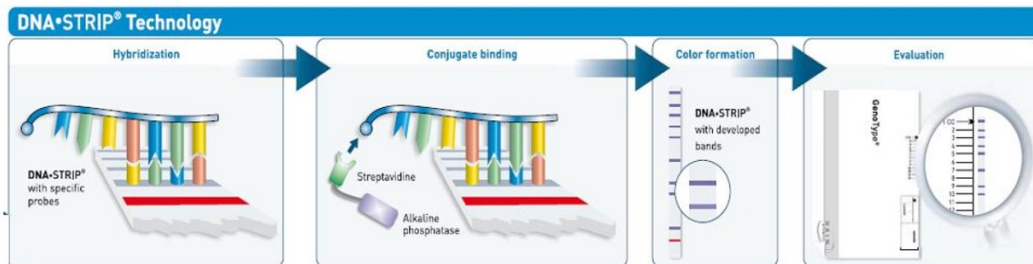
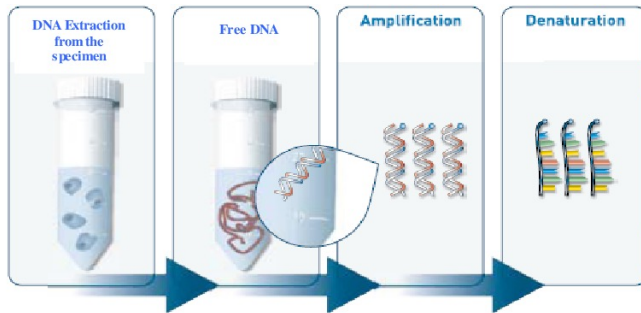
- Real-time PCR: GeneXpert^R (MTB/RIF)
 - Identify *M. tuberculosis* complex (MTBC)
 - Detect Rifampicin resistance gene (*rpoB* gene)
 - WHO recommendation





Molecular diagnostic method

- PCR/reverse hybridization: Line Probe Assay (LPA)
 - Identify MTBC and antibiotic-resistant genes, including first-line drugs and second-line drugs
 - Differentiate NTM species

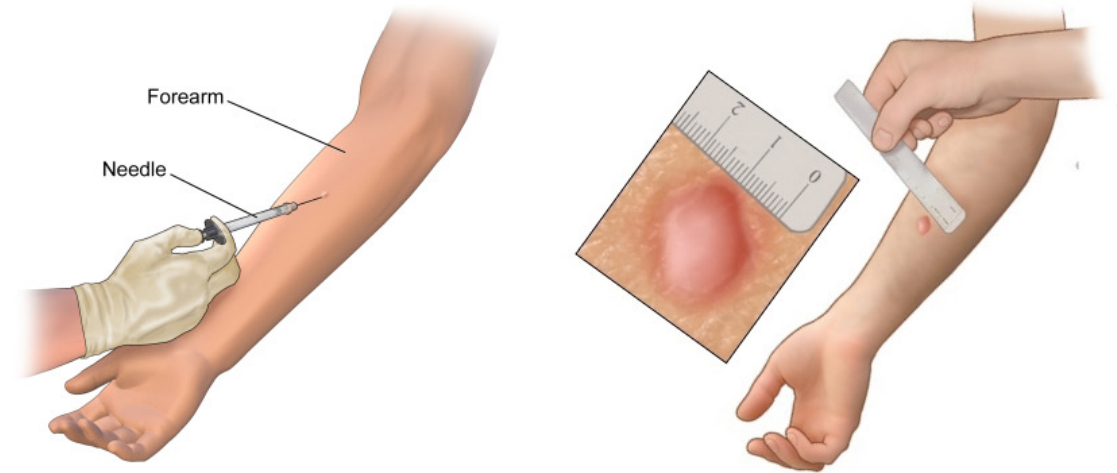




Immunodiagnostic testing

• Tuberculin skin test

- Assess patient's response to *M. tuberculosis* exposure
- Used Purified Protein Derivative (PPD) of mycobacterial antigen for injection at dorsal surface of the forearm
- Read the result after 48-72 hours by measured in millimeters of induration
- **False negative : NTM infection**
BCG vaccination





Immunodiagnostic testing

- Interferon-gamma release assay (IGRA)

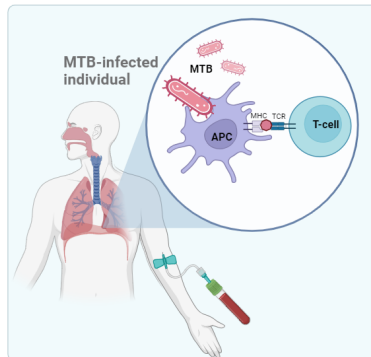
- Measuring the release of interferon-gamma (IFN- γ) from T-cells upon exposure to specific mycobacterial proteins
 - ESAT-6
 - CFD-10
- Cannot distinguish latent TB from active TB

Steps of Interferon gamma release assay (IGRA)

1

Blood Collection

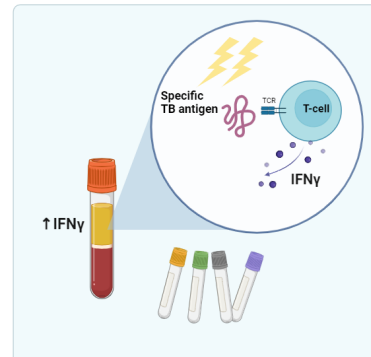
Collect a blood sample from the individual



2

T-cell Stimulation

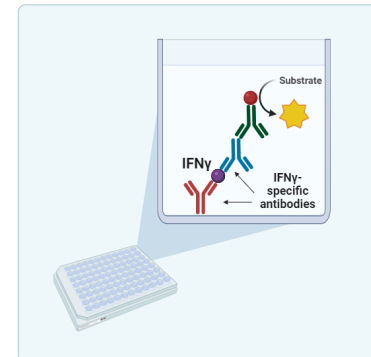
Stimulate T-cells with TB-specific antigens to trigger the release of IFN γ in antigen-coated tubes



3

IFN γ Measurement

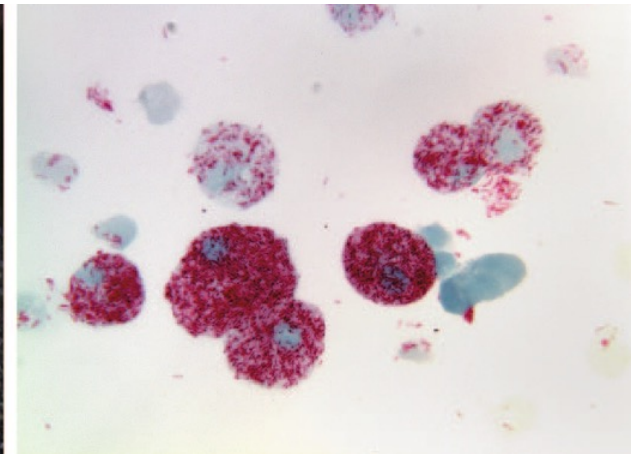
Measure IFN γ release using ELISA





Lepromatous Mycobacteria

- Non-cultivable in artificial media
- Grow only in specific animal models (mouse footpads, armadillos)
- Doubling time : 10-14 days
- Contains 2 species : *M. leprae*, and *M. lepromatosis*
- Mostly found in warm tropical countries
 - Endemic area: Mexico
- Transmission: respiratory droplets or skin lesions





Clinical manifestations

- Two forms

- Tuberculoid Leprosy

- Localized
- Hypopigmented, **anesthetic** skin lesions with well-defined edges
- **Low bacterial load (Paucibacillary)**
- **Peripheral nerve thickening and sensory loss**
- **Th1-dominant**

โรคผิวหนัง มีอาการไม่พบ AFB

ผิวหนัง ๒

Peripheral nerve เพิ่มขึ้น และสูญเสีย sensory



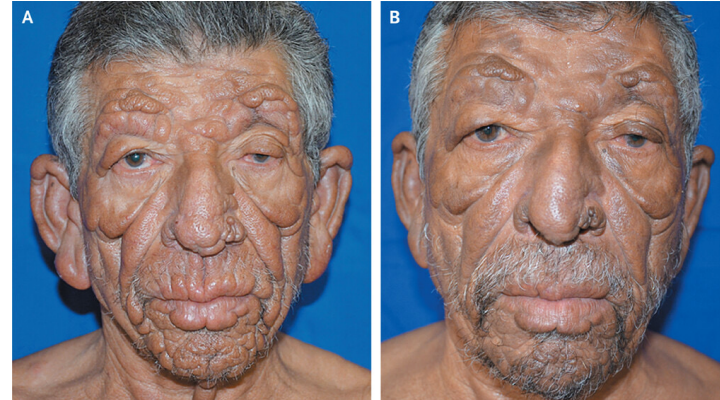


Clinical manifestations

- Two forms

- Lepromatous Leprosy

- Disseminated ไม่จำกัด ที่ผิวหนัง
- Symmetric, nodular skin lesions, นูนขึ้น thickened dermis, and widespread nerve damage ทำให้อาการเกี่ยวกับ nerve
- High bacterial load (multibacillary)
- Weak cell-mediated immunity and Th-2 dominant



leonine facies



deformities

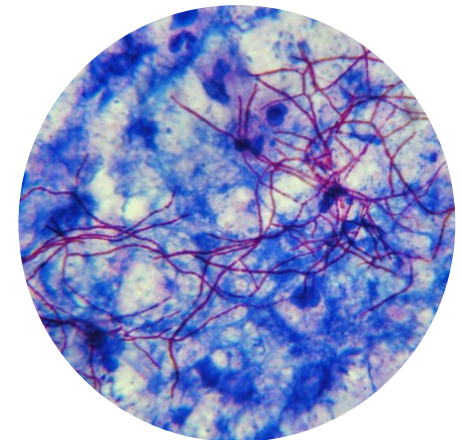
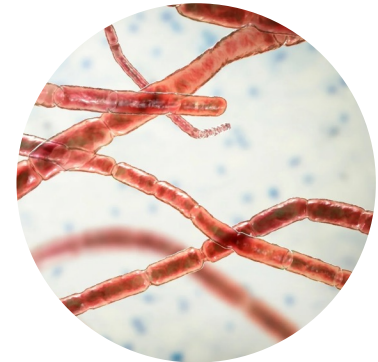


Treatment

- Tuberculoid Leprosy
 - Dapsone (anti-leprosy agent)
 - Rifampin
 - Duration: 6 months
- Lepromatous Leprosy
 - Dapsone (anti-leprosy agent)
 - Rifampin
 - Clofazimine
 - Duration: 12 months



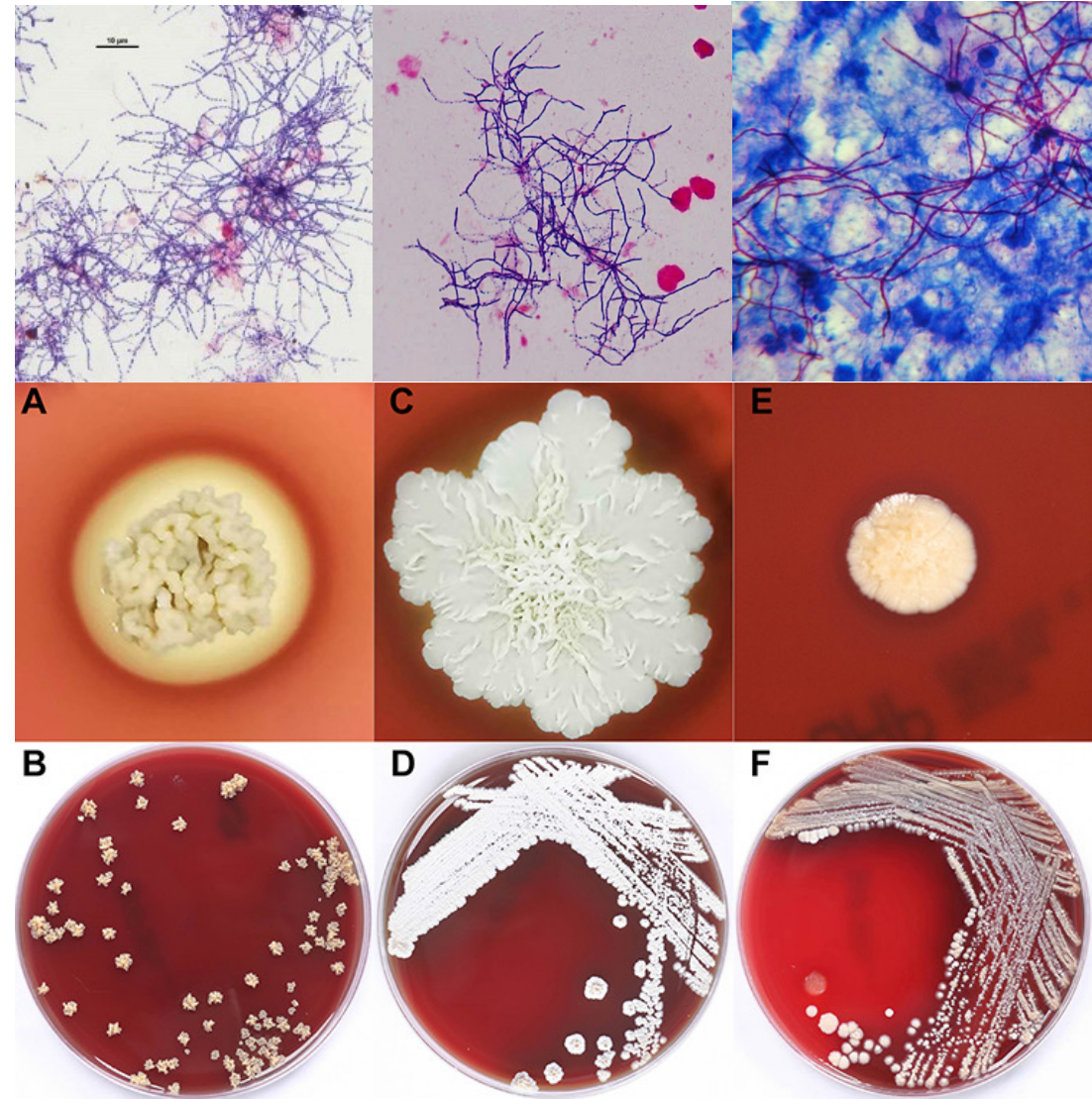
- Belong to the order Mycobacteriales, family Nocardiaceae
- Strict aerobic rods that form branched filaments in tissues and culture
- More than 100 species have been identified
- Cell wall contains mycolic acid (lipid-rich cell wall)
- Opportunistic pathogen
- The length of the mycolic acids: 50-62 atoms
 - Partially acid fastness (weakly acid fast) --> modified AFB positive
 - Gram's stain --> gram-positive bacilli with branching filament
- Can be culture on most nonselective laboratory media
 - Blood agar (BA)
 - Slow growth (required 3-5 days for incubation)





Human pathogenic strains

- *N. asteroides* complex
 - *N. farcinica*
 - *N. cyriacigeorgica*
 - *N. abscessus*
 - *N. brevicatena*
 - *N. nova*
 - *N. transvalensis*
- *N. beijingensis*
- *N. otitidiscaviarum*





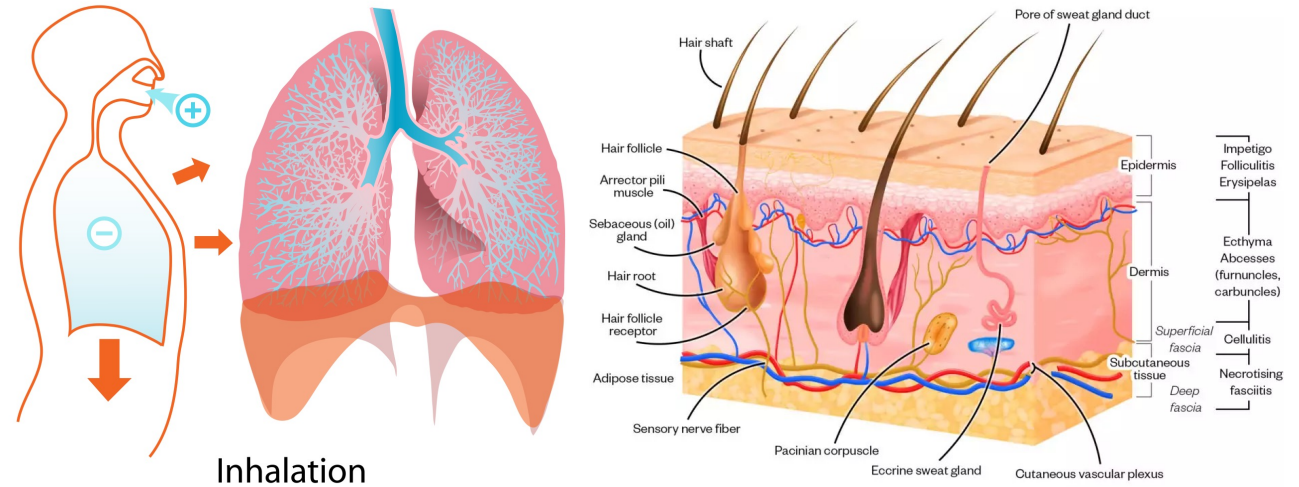
Epidemiology and risk factors

- Normal inhabitants of soil, water, decaying vegetation, and decaying organic matter
- In US, 800-1,000 cases per year
- Increasing incidence due to an increasing population of immunocompromised individuals worldwide.
- More common in males than females in a 3 to 1 ratio
- 60 % of cases are immunocompromised individuals

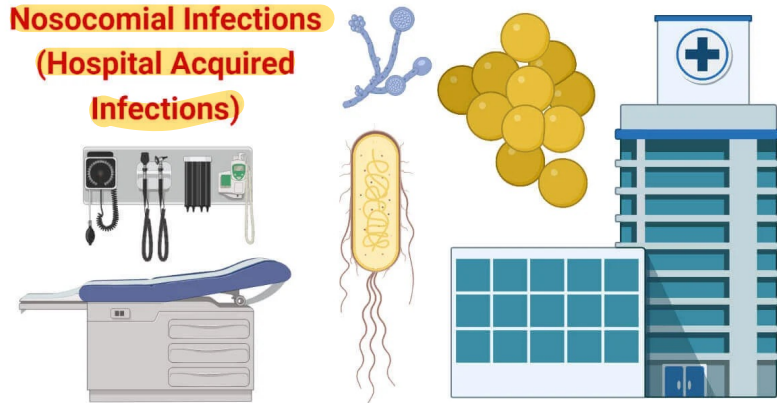


Transmission

- **Inhalation (Primary route)**
- Direct traumatic inoculation
- **Hospitalized acquired infection**



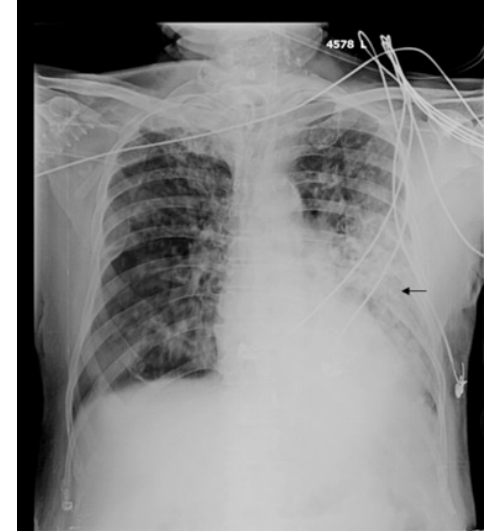
Nosocomial Infections (Hospital Acquired Infections)





Clinical Manifestations

- Bronchopulmonary disease (Pulmonary Nocardiosis)
 - Cannot be distinguished from other pyogenic organisms
 - Commonly found in immunocompromised patients
 - Cavitation and spread into the pleura
 - May be disseminated to CNS / subcutaneous tissue (50%)
- Cutaneous infections
 - Mycetoma, lymphocutaneous, cellulitis, subcutaneous abscesses
 - Secondary infection from pulmonary infection
 - May produce sulfur granules on skin lesions
- Central nervous system
 - One-third of all patients
 - Formation of single or multiple brain abscesses



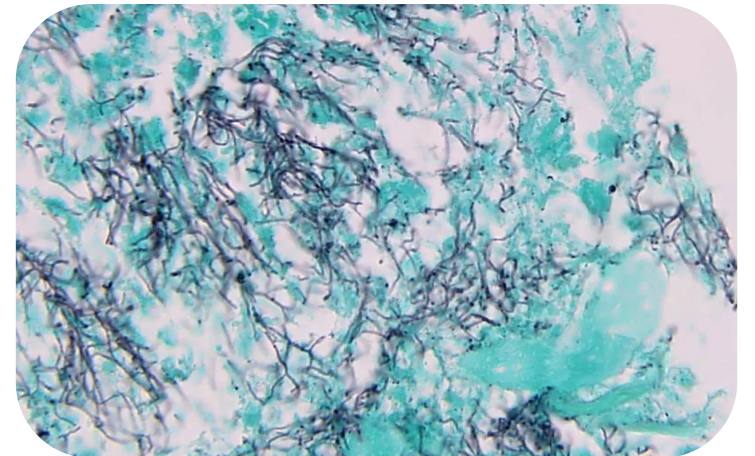
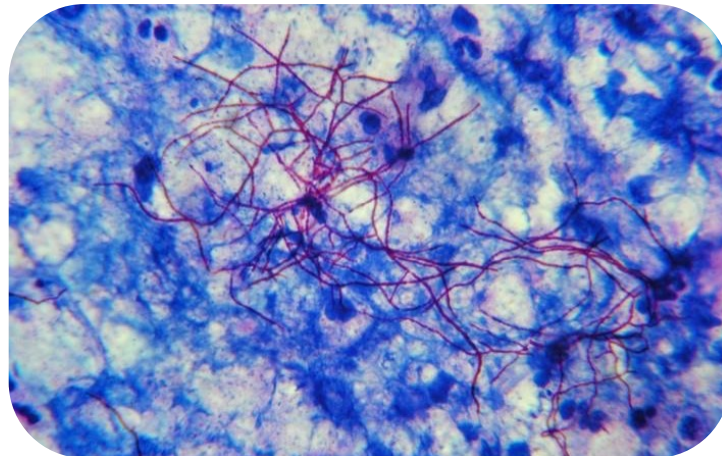
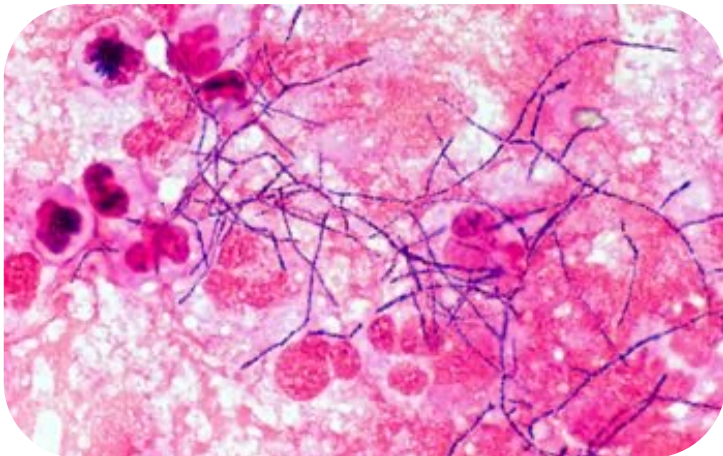
Sulfur granule



Laboratory diagnosis

- Direct microscopic examinations
 - Gram stain : Gram-positive bacilli, beaded, branching filament
 - Modified acid fast bacilli (mAFB) stain : Positive with branching filament
 - Histopathological stain

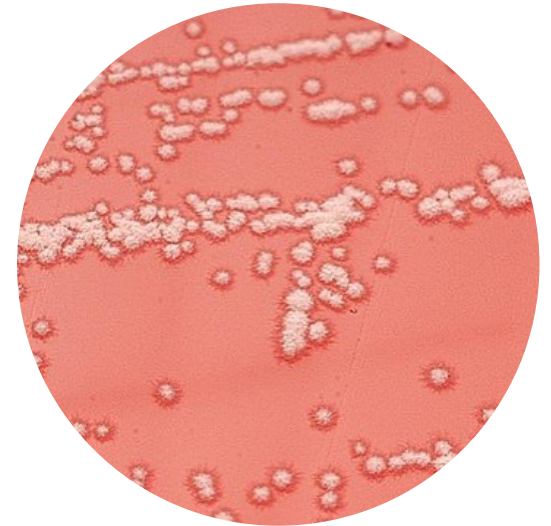
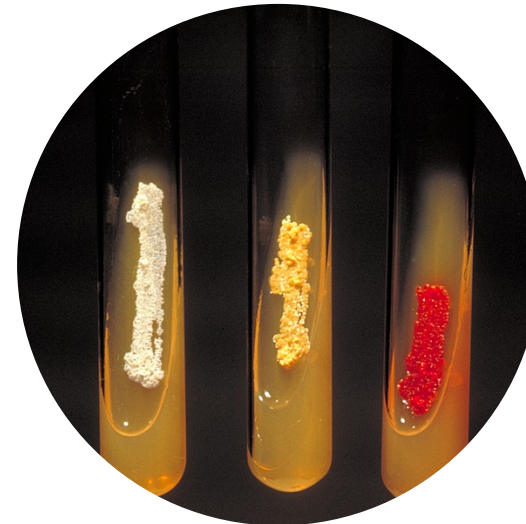
Decolorizer : Acid alcohol --> 1% H₂SO₄ in 95% ethanol
 - Gomori's methenamine silver (GMS) stain: found filamentous bacteria





Laboratory diagnosis

- Cultivation
 - Gold standard method
 - Growth on routine laboratory media
 - Blood agar, Potato Dextrose Agar (PDA)
 - Colony morphology : Extremely variable
 - Often in "Chalky-white"
 - Aerial hyphae
 - adherent into agar medium
- Incubation time: 2-7 days





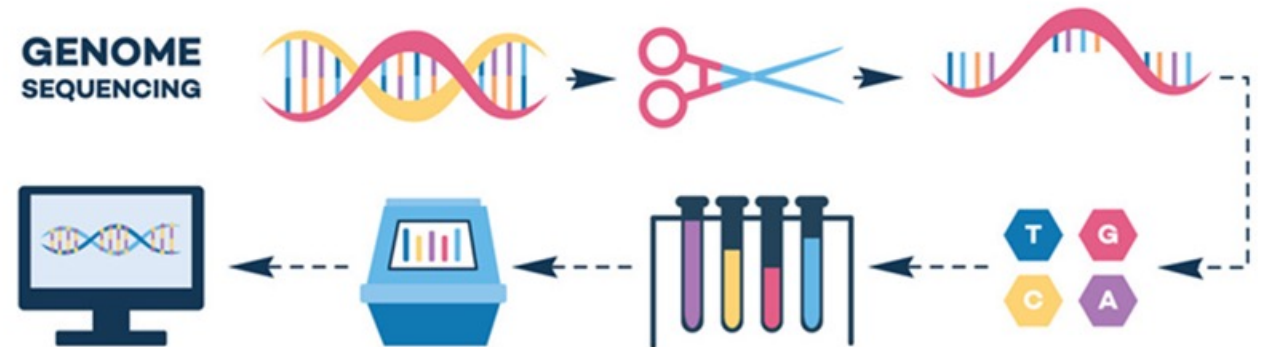
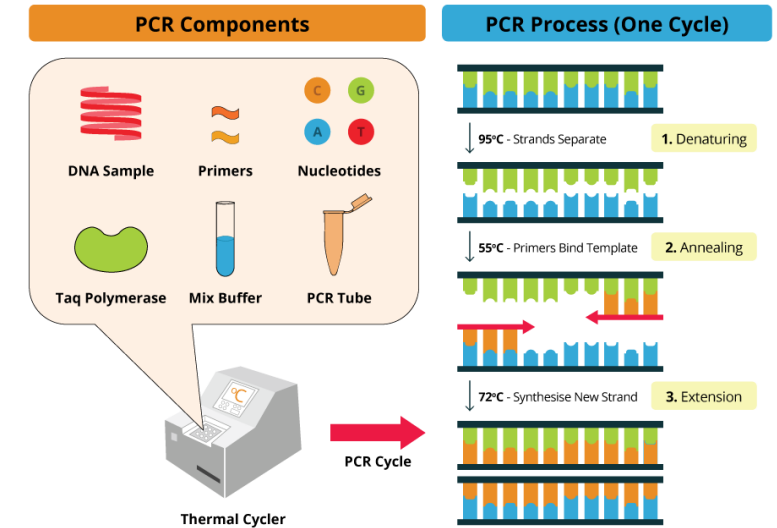
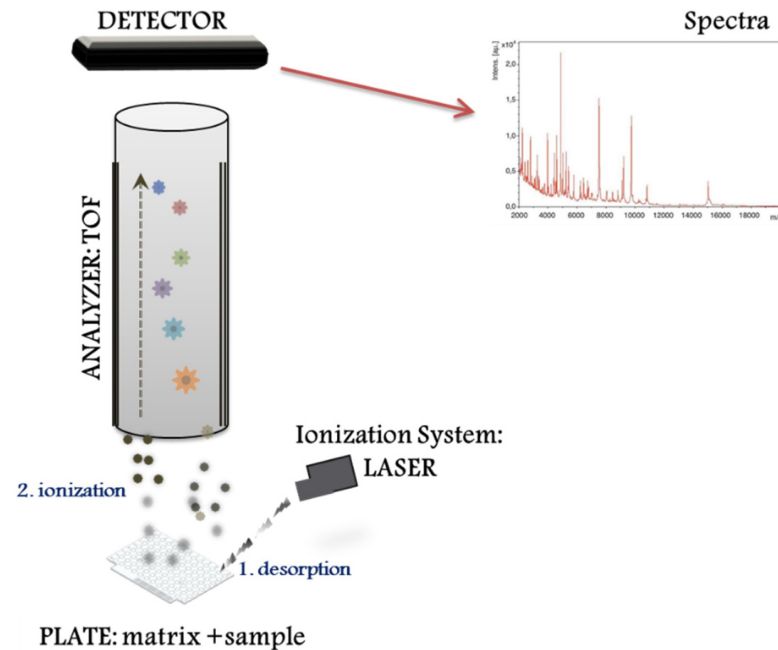
- Laboratory diagnosis

- MALDI-TOF** *m/z* PATTERN vs Protein

- Proteomic profile

- Molecular techniques**

- Genomic profile: PCR, genome sequencing





Treatment

- **Sulfonamide** : trimethoprim-sulfamethoxazole
 - Noted : several species have been shown resistant phenomenon
- **Recommended**
 - Trimethoprim-sulfamethoxazole + imipenem or amikacin
- **CNS involvement**
 - Trimethoprim-sulfamethoxazole + ceftriaxone
- **Future**
 - Oxazolidinone or linezolid
- **Duration**
 - Immunocompetent : 6 – 12 months
 - CNS involvement or immunocompromised : > 12 months



Rhodococcus spp.

- Belong to the order Mycobacteriales, family Nocardiaceae
- Strictly aerobic gram-positive coccobacilli or short bacilli (pleomorphic)
- Clinically important strain : *Rhodococcus equi*
equi = equine = Horse
- Facultative intracellular pathogen
- Zoonotic pathogen
- Cell wall contains mycolic acid (lipid-rich cell wall)
- The length of the mycolic acids: 30-54 atoms
 - Partially acid fastness (weakly acid fast) --> modified AFB positive
- Can be culture on most nonselective laboratory media
 - Blood agar (BA)
 - Slow growth (required 3-5 days for incubation)





Epidemiology and risk factors

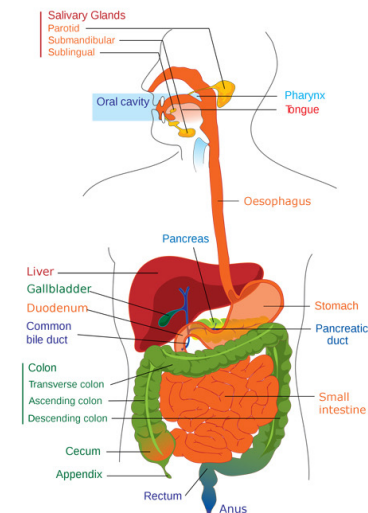
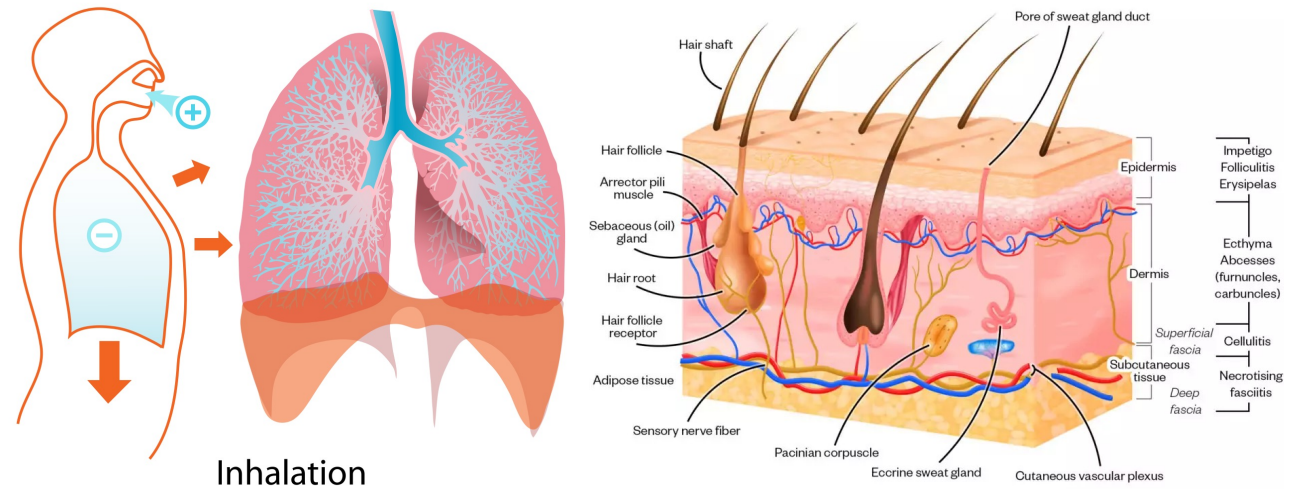
- Normal inhabitants of soil, water, air and plant surfaces
- Gastrointestinal tract of numerous herbivores
- Associated with occupational disease in farmers and veterinarians
 - Contact with grazing animals
 - Expose to soil and water contaminated by herbivore manure
- An increased number of cases involving immunocompromised patients
 - Nosocomial and catheter-associated infections
- Virulence
 - Resistance to intracellular killing
 - Ability to inhibit phagosome-lysosome fusion
 - Production of catalase, superoxide dismutase
 - Virulence-associated protein A (*vapA*) – highly virulent strain
 - Virulence-associated protein B (*vapB*) – intermediate virulent strain
 - Lacking virulence-associated plasmid (*vap*) genes – avirulent strain (50% of immunocompromised individuals)





Transmission

- Inhalation (primary route)
- Direct traumatic inoculation
- Ingestion





Clinical Manifestations

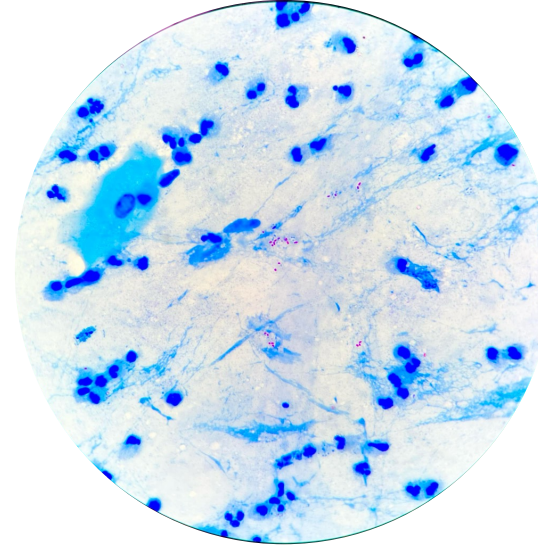
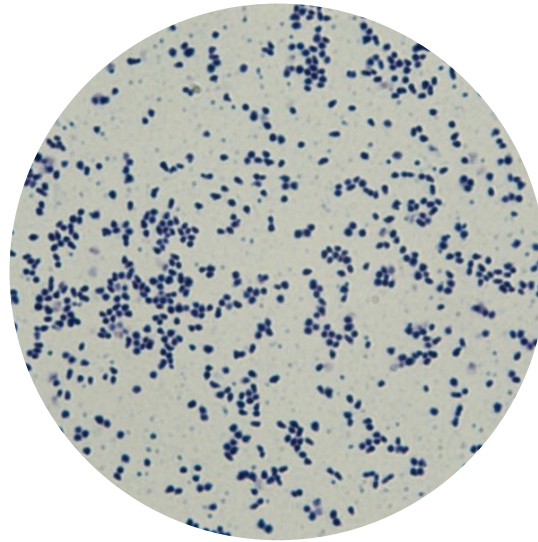
- Pulmonary infections (80%)
 - Mainly found in immunocompromised patients
 - Lung abscess
- Extrapulmonary infection (~20%)
 - Bacteremia
 - Skin, urinary tract, and wound infections
 - Endophthalmitis
 - Peritonitis
 - Catheter-associated sepsis
 - Abscesses
 - Osteomyelitis





Laboratory diagnosis

- Direct microscopic examinations
 - Gram stain : Gram-positive cocci, coccobacilli, short bacilli
 - Modified acid fast bacilli (mAFB) stain : Positive with coccobacilli shape

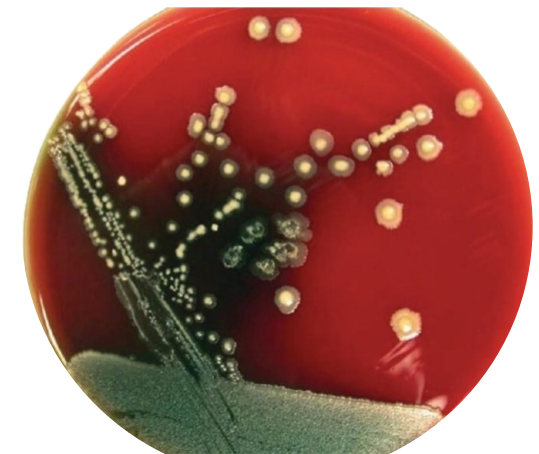
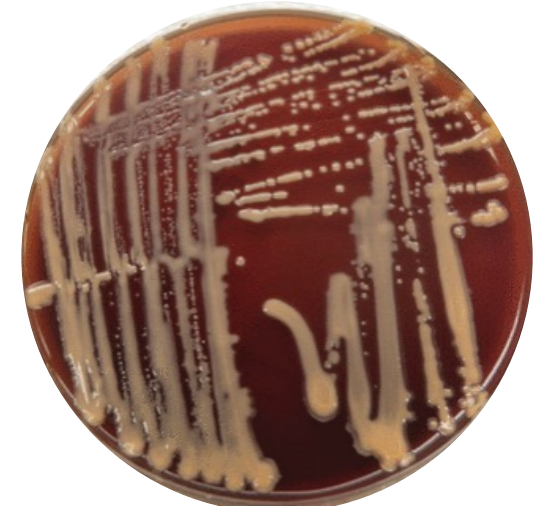




Laboratory diagnosis

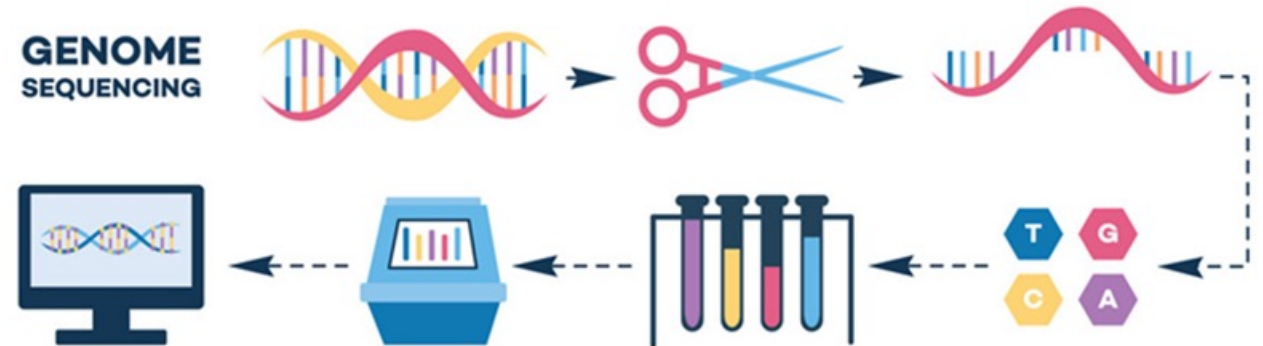
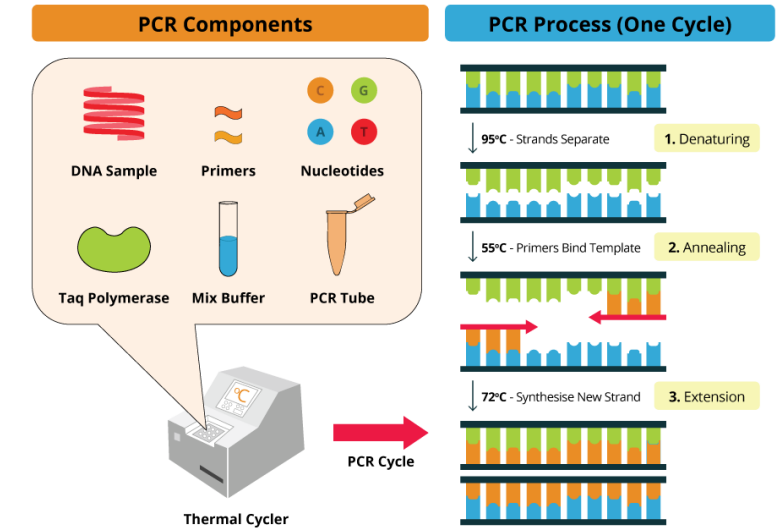
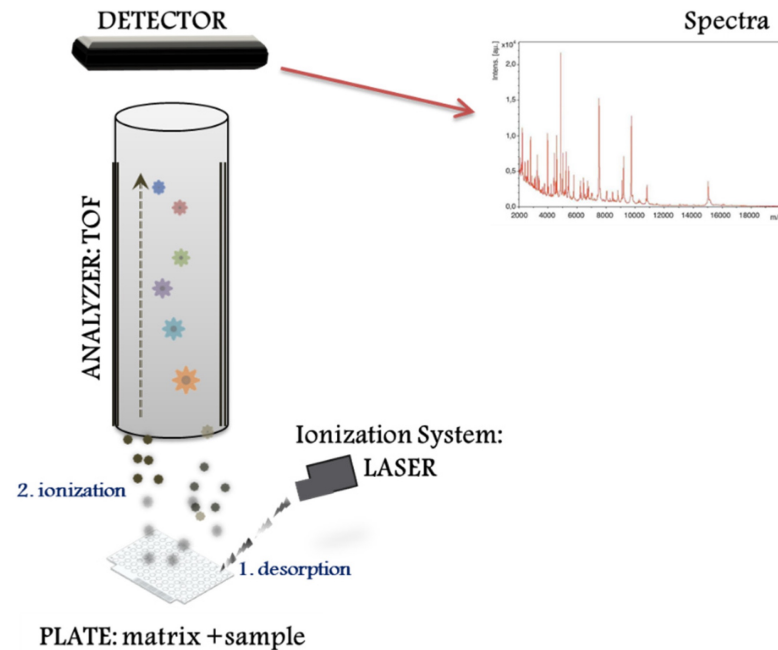
- Cultivation

- Gold standard method
- Growth on routine laboratory media
 - Colony morphology : rough, smooth or mucoid
 - Color : cream, yellow, coral, orange
 - "Salmon-pink pigment" can be observed within 4-7days





- Laboratory diagnosis
 - MALDI-TOF
 - Proteomic profile
 - Molecular techniques
 - Genomic profile: PCR, genome sequencing





Treatment

- **Beta-lactams** (except carbapenem) should be **avoided**
- **Immunocompetent host**
 - Single agent : extended-spectrum macrolide
fluoroquinolones
- **Immunocompromised host**
 - Combination agent : at least 1 demonstrating good macrophage penetration
 - Macrolides – azithromycin, clarithromycin, erythromycin
 - Fluoroquinolones – levofloxacin, ciprofloxacin
 - Tetracyclines – doxycycline, minocycline
 - Rifamycins – rifampin, rifabutin**
- **Duration**
 - 2 – 6 months



Aerobic *Actinomycetes*

- *Mycobacterium* spp. Acid-fastness
- *Nocardia* spp. Partially acid-fastness
- *Rhodococcus* spp. Partially acid-fastness
- *Gordonia* spp. Partially acid-fastness
- *Tsukamurella* spp. Partially acid-fastness
- *Dietzia* spp.
- *Streptomyces* spp.
- *Actinomadura* spp.
- *Tropheryma whipplei*



- Closely related to *Rhodococcus* spp.
- Slow-growing bacteria (2-7 days)
- Partially acid fastness (contain mycolic acid) = modified acid-fast stain +
- Opportunistic infection in human
 - *Gordonia* spp. : skin infections, chronic pulmonary disease, catheter-associated sepsis, bacteremia, central nervous system
 - *Tsukamurella* spp. : peritonitis, catheter-associated sepsis, skin infection, meningitis



Streptomyces spp.

- The largest group in phylum *Actinomycetota*
 - More than 3,000 species
 - Mostly distributed in soil and produce spore
- Ability to synthesize > 50 different antibiotics
 - E.g. streptomycin, lincomycin, daptomycin, chloramphenicol
- There are few strains associated with human disease
 - *S. somaliensis* and *S. sudanensis*
- Strictly aerobic gram-positive branched mycelium with rarely filamented

